NG1000 Weather Radio Transmitter System

Operations and Maintenance Manual

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nautel

Nautel Limited
10089 Peggy's Cove Road
Hackett's Cove, NS Canada B3Z 3J4
Phone: +1.902.823.3900 or
Toll Free: +1.877.6NAUTEL (6628835) (Canada & USA only)
Fax: +1.902.823.3183

Nautel Inc.
201 Target Industrial Circle
Bangor, Maine USA 04401
Phone: +1.207.947.8200
Fax: +1.207.947.3693

Customer Service (24 hour support)
+1.877.628.8353 (Canada & USA only)
+1.902.823.5100 (International)

Email: support@nautel.com
Web: www.nautel.com

The comparisons and other information provided in this document have been prepared in good faith based on publicly available information. The reader is encouraged to consult the respective manufacturer's most recent published data for verification.

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Release control record

Issue	Date	Reason
0.1	2010-02-24	Initial release of manual
0.2	2010-05-19	Incorporated Engineering review coments; updated front panel screen shots
0.3	2010-06-11	Incorporated Engineering review comments; added NG Series Remote Application

About this manual

This manual provides technical information needed when operating, maintaining and troubleshooting an NG1000 transmitter. This manual is intended for use by transmitter operators and field technicians.

Using this manual

If you are responsible for configuring or operating a transmitter, see Section 2, "Operating the transmitter" on page 2-1.

If you are performing scheduled maintenance, or planning your maintenance schedule, see Section 3, "Routine maintenance" on page 3-1.

Performing procedures

When using procedures in this manual, perform each step in sequence.

- If you are asked to **see** another section of this manual, or another document, refer to that section or document for additional information, then continue the procedure.
- If you are asked to **go** to another step within the procedure, jump directly to that step without performing the intervening steps.
- If you are asked to **go** to another section or document, stop the procedure and perform the tasks described in the other section or document.
- If you are asked to **check a voltage**, use a digital voltmeter and test the voltage relative to ground (unless otherwise instructed).
- If you are asked to **check a signal**, use an oscilloscope and test the signal relative to ground (unless otherwise instructed).

CAUTION: FAILURE TO COMPLY WITH RECOMMENDATIONS MAY VOID YOUR MANUFACTURER'S WARRANTY. FOR MORE INFORMATION, REVIEW YOUR WARRANTY DOCUMENTS.

Tip: When you have completed a task or a step, put a checkmark beside the step number.

Technical support

Nautel offers technical support to customers over the Internet and by telephone. Nautel's customer support team will answer your questions and work with you to identify and resolve problems.

For 24-hour technical support, call toll free at 1.877.628.8353 (in USA and Canada only) or call 1.902.823.5100 (international) or find us on the Internet at http://www.nautel.com.

For parts and tools information, see "Parts and tools" on page 9-1 of the *NG1000 Pre-Installation Manual*.

For extended warranty information, see "Pre-installation assistance" on page 9-1 of the *NG1000 Pre-Installation Manual*.

NG1000 transmitter manuals

The NG1000 documentation suite includes the following documents:

NG1000 Pre-installation Manual, NG1000-PREINST. Provides instructions and reference information needed when planning and preparing for the installation of an NG1000 transmitter.

Nautel Site Protection Manual. Provides detailed information about protecting your site from lightning-related hazards.

NG1000 Installation Manual, NG1000-INST. Provides instructions and reference information needed when installing an NG1000 transmitter.

NG1000 Operating and Maintenance Manual, NG1000-OPS-MAINT. Provides instructions for operating, maintaining and troubleshooting an NG1000 transmitter. It also provides reference information needed when performing diagnostic procedures.

NG1000 Troubleshooting Manual, NG1000-TROUBLE. Provides detailed technical information about the NG1000 transmitter, including electrical schematics and mechanical drawings.

Nautel website / Online resources

The Nautel website provides useful resources to keep you up to date on your NG1000.

Nautel User Group (NUG)

The website includes a special section that customers can log into in order to access the Nautel customer newsletter, product manuals, frequently asked questions (FAQ), information sheets, and information about field upgrades.

Documentation: online and printed

The website's NUG section provides online access to all the documentation for your NG1000. Documentation is provided in Acrobat (PDF) format. You can use the documentation online or print the sections that you need.

When using online documents:

- Click on blue text (hyperlinks) to jump to a related section, or to get additional information (e.g., view a term's definition).
- To search a document to find keywords, use Find in Acrobat Reader's Edit menu.
- To quickly find a specific section, click the section in the PDF file's Bookmarks list.

When using printed documents:

- To find keywords, go to the *Index* section at the end of the manual.
- To find a specific term, go to the *List of Terms* section near the end of the manual.

About safety

All Nautel transmitters are designed to meet the requirements of *EN60215, Safety Requirements for Radio Transmitters*.

The philosophy of *EN60215* is that the removal of any cover or panel that can only be opened using a tool is a maintenance activity, and that any person performing a maintenance activity is expected to be trained for that activity. Under *EN60215*, it is assumed that trained personnel will be knowledgeable and will take precautions such as removing all power to the transmitter before accessing its components.

Electrical hazards

To remove power from the transmitter, switch off and lock out the ac power. There is an amber LED in each ac distribution assembly at the bottom rear of the cabinet that glows to remind anyone who has not turned off the power that the system is live and serious danger is present.

WARNING: It is not enough to remove RF power. The power line is still connected.

Mount the transmitter ac power disconnect switch/breaker close to the transmitter so that it can be reached quickly in an emergency. Clearly label the disconnect switch/breaker (e.g., EMERGENCY SWITCH).

After turning off the power, always perform a measurement to confirm that the power is off before touching anything within the transmitter. If the wrong breaker was opened, the equipment will be live.

WARNING: Do not use an ordinary multimeter to check for voltage, since it may have been left inadvertently on the AMP (A) range, triggering a short and an arc blast that could result in severe burns and even death.

Use only a non-contact voltage probe or a safety voltmeter (available from vendors such as Fluke, Ideal, and Teagam).

Use a proper lockout procedure to ensure that another worker cannot accidentally reapply power while you are performing maintenance on any part of the transmitter or site.

Lightning hazards

Before opening the transmitter and touching internal parts, remove and solidly ground the antenna connection.

WARNING: It is not enough to ground the antenna terminal with the antenna still connected. Even a small impedance in the ground strap will result in lethal voltages during a lightning strike.

RF hazards

A serious RF hazard and very high voltages exist in the vicinity of the antenna and its networks during normal operations.

Toxic hazards

There are devices used in this equipment containing beryllium oxide ceramic, which is non-hazardous during normal device operation and under normal device failure conditions. These devices are specifically identified with "(BeO)" in the *Description* column of the *Troubleshooting Manual's* parts list(s).

Do not cut, crush or grind devices because the resulting dust may be hazardous if inhaled. Unserviceable devices should be disposed of as harmful waste.

Other hazards

Ensure that appropriate fire alarms and fire extinguishers are available. Extinguishers must be suitable for use on electrical fires.

Many other site safety risks exist. It is beyond the scope of this manual to identify all the risks and procedures.

Safety precautions

This section provides very important information about protecting the safety of personnel and equipment:

- Personal safety - see page xv
- Site safety - see page xvi
- Equipment safety - see page xviii

Personal safety

Training

The training of any personnel who will have physical access to the site or the transmitter is very important. Personnel must be familiar with the transmitter, so that they can avoid physical danger, and be aware of hazards to themselves and the equipment.

Nautel offers a number of training courses covering the basic fundamentals of RF systems and transmitters, and the operation and maintenance of the transmitter. For more information about available courses and schedules, go to the Nautel website at http://www.nautel.com/Training.aspx, or ask your Nautel sales representative.

Site orientation

When you give personnel access to the transmitter site (e.g., hiring new personnel, or giving access keys to personnel), perform a site orientation to ensure that they are familiar with the site, on-site procedures, and on-site hazards. Cover the following topics:

- Securing the site (locking doors and fences) to prevent unauthorized access
- How and when to call for technical support or emergency assistance
- Areas of the site and pieces of equipment that are *off limits*

Voltage awareness

Ensure that all personnel that are able to access areas with high voltage circuits or high field strengths are aware of the hazards associated with high voltage. Cover the following topics:

- High voltage or high field strength areas where caution is required
- Physical risks of electric shock
- Risks for personnel with pacemakers or other medical implants
- Induced voltages in high field strength areas
- On-site risks during thunderstorms and lightning strikes
- Operation of safety interlocks (if installed)

First aid

Nautel does not offer first aid training, since the hazards associated with high voltage and RF energy are not specific to the transmitter. However, the customer should provide first aid training to all personnel who have access to the transmitter site. First aid training should include CPR, care of burns, artificial respiration, and defibrillation if specific equipment is available on-site.

Site safety

Controlling access

Transmitters and antennas generate and carry dangerous voltages that can be harmful or fatal. It is very important that you control access to the site and its equipment. To secure your transmitter site, use:

- Locking steel or security doors to prevent casual access
- A perimeter fence to keep trespassers away from the antenna system and feedline
- “No Trespassing” signs
- An alarm system

Marking hazards

Place warning signs close to any hazardous areas or systems (e.g., the feedline or the antenna system). Make the signs large enough that they cannot be missed. Provide signage in all languages used in the region. These signs are intended not only for authorized personnel, but also for emergency responders or accidental trespassers.

Qualifying site personnel

Make sure that personnel who have access to the site are qualified to work around electronics and high voltage systems.

Ac power protection

You should take steps to protect equipment from surges (over-voltage spikes) on the ac power lines. Surges may occur during thunderstorms, or because of malfunctions in the electrical distribution grid. Surge suppressors and ac power conditioners can prevent serious damage to your on-site equipment, including the transmitter.

RF protection

Transmitters and their antenna systems create intense radio frequency fields at the transmitter site, particularly near the feedline, antenna and tower. At some sites, these fields may cause biological effects, including the heating of body tissues. Intense fields can also create dangerous high voltages on ungrounded, conductive surfaces and objects. At certain points where high voltage conductors come close to grounded conductors (e.g., at feedline junctions or on the tower), dangerous electrical arcing or flashovers can occur. It is very important that you take the following steps to prevent damage to equipment or personnel due to RF fields:

- Use safety interlocks to de-energize transmitters if personnel open doors or panels accessing high field areas
- Place warning signs in any locations where high fields can occur
- Train personnel about the short-term and long-term hazards of RF radiation
- Physically block access to the area around the antenna system, feedline and tower
- Ground all exposed conductive surfaces or objects in high field areas

The RF connection to the transmitter output can be a serious safety hazard. Connect a 50 Ω test load during installation and commissioning. It is recommended that a switch be used to automatically connect the transmitter to the antenna system without human contact with the transmitting conductors.

Safety interlocks

The transmitter contains an electrical interlock, which is an external circuit that turns off the RF output if any of its switches are opened.

Ac disconnect switch

Safe operation of the transmitter requires an ac disconnect switch. Lock the ac disconnect switch in the disconnected (open) position during the installation process.

Equipment safety

Electrostatic protection

The transmitter's systems are very rugged and resistant to damage. However, it is possible for damage to occur because of high voltage electrostatic discharges during servicing. Train all service personnel to ground themselves to bleed off any static charge before opening the transmitter or touching any exposed components. Provide a grounding wand or known ground (e.g., a grounded metal table) that personnel can use to discharge themselves.

Surge protection

Surge protection is recommended for your entire site. However, even if you do not use a surge protector on the service entrance to the site, you should install a surge protector in the transmitter's ac power feed to prevent over-voltage from entering the transmitter.

Lightning protection

The transmitter is designed to resist lightning strike damage. However, intense or repeated strikes could damage the transmitter. We recommend that you install lightning suppression on the antenna, tower and feedline to reduce the effect of lightning strikes on the transmitter itself (and to protect the rest of your site equipment and your personnel). For detailed information about lightning protection, see the *Nautel Site Preparation Manual*, available from your Nautel sales agent, or online from the Nautel website.

Physical protection

Consider physical hazards to equipment at your site, including the transmitter. Ensure that equipment is protected from weather (e.g., rain or flooding), even during extreme weather events. Place equipment so that it is not in the path of swinging doors or high-traffic areas. Do not allow wheeled items like office chairs or tables with wheels in the transmitter room, as these may damage equipment if accidentally pushed or knocked over. Do not place the transmitter under water pipes, drains, or sprinklers. Keep any equipment that generates heat, like the transmitter, away from flammable materials like ceiling panels, cubicle dividers, and curtains.

Earthquake protection

If the transmitter site is in a region that experiences any noticeable earthquake activity, take steps to prevent the transmitter from shifting or rocking during an earthquake. Even during minor earthquakes, rocking or movement of the transmitter is likely to damage the feedline connection, and could even cause a catastrophic failure of the ac power feed into the transmitter. During larger earthquakes, the weight of the transmitter chassis could be hazardous to nearby equipment or personnel.

Four 1/4-inch pilot holes are provided in the bottom corners of the transmitter system cabinet to allow for securing the cabinet to a floor.

Section 1: Description

This section provides a high-level description of the transmitter system's key sections.

Some descriptions in this section refer to electrical schematics (SD-#s) . These are located in Section 5 of the *NG1000 Troubleshooting Manual*.

NG1000 Weather radio transmitter system

See the block diagram in Figure 1.1 on page 1.7 and electrical schematics SD-1 and SD-2.

The NG1000 transmitter system is a 1000 W main/standby (primary/secondary) analog VHF transmitter operating in the 162.4 MHz to 162.55 MHz band. The system is intended for use as a weather radio transmitter that is capable of delivering a continuous stream of audio programming - in the form of weather messages - directly to the public. These messages may include:

- Informational climatological outlook
- Weather warnings that may pose immediate verified threats to the safety and well being of the public.
- Appended to these warnings are signals and digital codes that can alert individuals to specific threats in specific areas and that can be used to trigger the FCC's Emergency Alert System (EAS).

The NG1000 is housed in a cabinet that is 48 inches high, 23 inches wide and 30 inches deep (not including the external bandpass filter and dummy load). The NG1000 consists of the following components:

- Two 1250 W VHF transmitters (A and B, in a main-standby configuration)
- System controller - see page 1-6
- RF coaxial switch - see page 1-6
- Dummy load - see page 1-6
- Bandpass filter
- System output directional coupler
- Primary/secondary ac distribution units
- Audio surge protector
- Modem - see page 1-6
- ROAMS unit

1250 W VHF transmitters

See the block diagram in Figure 1.2 on page 1.8 and electrical schematics SD-2, SD-8 and SD-9.

Each transmitter [A2 (A) and A3 (B)] is a totally solid state, VHF, frequency modulated transmitter, capable of producing up to 1250 W. Each transmitter consists of the following primary components:

- Exciter/control PWB
- RF section - see page 1-3
- Power supply section - see page 1-5

Exciter/control PWB

See electrical schematics SD-8 through SD-14.

The exciter portion of the exciter/control PWB (A1) performs the following functions:

- accepts the audio input and generates the RF drive required to modulate the audio signal at the desired carrier frequency
- measures the audio input level
- receives a sub-audible input signal and modulates it on the RF output
- provides a 10 MHz calibration input
- provides the status of any exciter fault conditions

The control portion of the exciter/control PWB (A1) performs the following functions:

- monitors the current value of various parameters including cooling fan tachometer outputs, transmitter output forward and reflected power, IPA output forward power, heatsink temperature, intake temperature, all PA current levels, and PA power supply status signals
- measures and monitors the output voltage from the PA power supplies as well as the low voltage supplies (+15 V, -15 V, +5 V and +3.3 V)
- accepts an interlock signal from the system controller to ensure that the system's RF coaxial switch is correctly positioned before enabling transmitter turn-on
- generates the dc bias signals for the IPA and PAs

- generates the signals that enable the PA power supplyies and also control their output voltage
- provides the status of any controller fault conditions
- contains monitoring and protection circuits to prevent transmitter damage under adverse or fault conditions, including high speed SWR protection and interlock/muting
- an integral DSP performs all audio processing including filtering, AGC, SAME decoding and FM baseband modulation
- interfaces with the RF and power supply sections of the transmitter system to perform control and monitoring functions
- provides remote interfacing for the transmitter

RF section

See the block diagram in Figure 1.2 on page 1.8 and electrical schematics SD-9 and SD-16 through SD-18.

The RF section accepts the RF input from the exciter/control PWB and generates the final RF output. It consists of a pre-amplifier/IPA PWB (A8), a splitter PWB (A4), eight power amplifier PWBs [A9 (PA1) through A16 (PA8)], a combiner/LPF/probe PWB (A5), an interface PWB (A3), a combiner interface PWB (A6) and cooling fans B1 through B5.

RF Drive Splitter Circuit

See Figure SD-9. The splitter PWB (A4) contains circuitry that accepts the RF drive source from the exciter/control PWB, provides drive to the pre-amp/IPA PWB and provides eight equal RF drive signals for the RF power amplifiers.

The RF drive signal from the exciter/control PWB is applied to the splitter PWB, where it is applied to the pre-amp/IPA PWB (A8). The IPA output of the pre-amp/IPA output PWB is re-applied to the splitter PWB, via the combiner/LPF/probe PWB (A5), for splitting into eight equal RF drive signals to drive the power amplifier (PA) PWBs. The splitting circuitry on the splitter PWB is formed by micro-strip transmission lines of specific length and characteristic impedance. These transmission lines act to match the impedance between the RF drive input and the eight outputs.

The splitter PWB also provides an interface point for pre-amp/IPA PWB biasing signals and fan dc supplies from the exciter/control PWB and fan tachometer and temperature signals to the exciter/control PWB.

Pre-amp/IPA PWB

See Figures SD-9 and SD-17. The pre-amp/IPA PWB (A8) contains two stages of push-pull, RF power amplifiers, the last of which provides 150 W (nominal) of RF power as the IPA output power. The pre-amp RF drive, pre-amp and IPA voltage and pre-amp and IPA bias inputs are provided by the exciter/control PWB (A1). Cooling air for the power amplifiers is provided by fans B1 through B5.

The pre-amp/IPA PWB accepts the low level *RF Drive* input from the splitter PWB (A4). This input drives the first stage of amplifier, or pre-amplifier. Its RF output is controlled by the *Pre-Amp V* input from the PS distribution PWB (A2) (set at +48 V) and the *Pre-Amp Bias* input from the exciter/control PWB (A1). The output of this amplifier is applied to the second stage amplifier, or IPA. Its output is controlled by the *IPA V* (set at +48 V) input from the PS distribution PWB (A2) and *IPA Bias* input from the exciter/control PWB (A1). An IPA output sample is provided by circuitry on the combiner/LPF/probe PWB (A5) and the combiner interface PWB (A6).

PA PWBs

See Figures SD-9 and SD-18. Each power amplifier PWB (A9 through A16) contains an identical, push-pull, RF power amplifier (PA). Each PA is capable of providing 300 W of RF power in the weather radio band (162.4 to 162.55 MHz). The RF output is controlled by the PA V input from the control/ interface PWB (A1). Cooling air for the PA PWBs is provided by fans B1 through B5.

Combiner/LPF/probe PWB

See Figure SD-9. The combiner/LPF/probe PWB (A5) combines the RF outputs from the eight PAs to provide a single RF output. A low-pass filter circuit attenuates harmonics.

The combiner/LPF/probe PWB, in conjunction with the combiner interface PWB (A6), also contains three power probes to monitor RF signals. See also Figure SD-16.

The first probe monitors the IPA output signal. A short section of micro-strip transmission line is in close proximity to the IPA output from the pre-amp/IPA PWB. Due to the proximity and orientation to the micro-strip, an RF voltage representative of the IPA output signal is induced in a power probe circuit on the combiner interface PWB. A dc voltage, proportional to the IPA output level, is applied as the *IPA Pwr Sample* to the exciter/control PWB for monitoring and protection circuits.

The second probe monitors the final RF output signal. The output from the low-pass filter is connected to the final RF Output by a micro-strip transmission line. Due to the probe's proximity and orientation to this line, RF voltages representative of transmitter forward and reflected power are induced in respective probe circuits on the combiner interface PWB. The forward and reflected power outputs are applied as the *Fwd Pwr Sample* and *Refld Pwr Sample* to the exciter/control PWB for monitoring and protection purposes.

The third probe provides an RF Monitor output (J2, accessible on the rear panel of the transmitter), which is a nominal -37.5 dBc (3.3 V rms @ 1250 W). This signal is intended for use with a modulation monitor or spectrum analyzer.

Power supply section

See the block diagram in Figure 1.2 on page 1.8 and electrical schematics SD-1 and SD-8.

The power supply section converts the ac power source to a positive dc voltage (PA volts) for the transmitter's PA PWBs. The transmitter accepts a single-phase, 50/60 Hz ac input that is between 180 and 264 V ac (208 V ac nominal). The power supply section also provides +48 V for the pre-amp/IPA PWB, cooling fans and the low level dc voltages for the control/monitoring circuitry throughout the transmitter.

The power supply section consists of two PA power supplies (U3 and U4), +48 V power supply (U2) and a PS distribution PWB (A2).

PA power supply

See Figure SD-8. Power supply modules U3 and U4 convert the ac input voltage to a regulated dc supply (PA volts) for all eight PA PWBs. Each power supply module is rated for a nominal 2000 W and provides an output voltage between 7 and 53 V (typically 27 V). The modules regulate the output voltage based on a *PA Volts Control* input from the exciter/control PWB. Test points on the associated PS distribution PWB allow monitoring of module presence, ac input status, inhibit signal, control voltage, PS fail alarm, temperature fault alarm and PA currents. Each module has a built-in cooling fan and each senses out-of-regulation and excessive temperature conditions and applies *PS Fail* and *PS Temp* alarm signals to the exciter/control PWB. Both conditions cause the power supply to shut itself down, thus reducing the transmitter's RF output.

+48 V power supply

See Figure SD-8. +48 V power supply module U2 converts the ac input voltage to a regulated +48 V dc supply for both the pre-amp/IPA PWB, cooling fans and the PS distribution PWB.

PS distribution PWB

See Figures SD-8 and SD-15. The PS distribution PWB (A2) interfaces between the dc power supplies (PA and +48 V) and the rest of the transmitter. It distributes the PA volts outputs to the PA PWBs, +48 V outputs to the pre-amp/IPA PWB and the cooling fans. The PS distribution PWB also contains dc-dc converters that use the +48 V supply to create +5 and +15 V supplies for the exciter/control PWB.

Dc current samples of each of the RF power supplies (PA volts, pre-amp volts and IPA volts) are created on the PS disribution PWB and provided to the exciter/control PWB over an SPI bus.

System controller

The system controller provides system control and monitoring functions for two 1250 W VHF transmitters connected in a main-standby configuration. The transmitters provide data to the system controller via an RS-232 serial interface. System control and monitoring (transmitter changeover via a coaxial switch and interlocks) are hardwired to the system controller.

The system controller performs the following functions:

- provides a user interface (UI) for the transmitter system
- maintains a 254-entry event log that records transmitter systems events (e.g., ac loss, high SWR, transmitter A/B events, etc.) by date and time
- performs output power control
- displays alarms, meter readings from the transmitters

The system controller consists of three functional blocks - control/display, remote interface and power supply (see Figure SD-1).

RF coaxial switch

An RF coaxial transfer switch (U1) provides a means to route the RF output of the active 1250 W VHF transmitter to the external antenna or the dummy load (U3). The system controller provides control signals to determine the position of the switch (position 1 = transmitter A to antenna; position 2 = transmitter B to antenna). Switch position status is reported back to the system controller to ensure switch function is normal.

Dummy load

The dummy load, rated for 1,500 W, accepts the RF output of the system's standby transmitter.

Modem

A modem (U5) is provided as an option for remote control and monitoring of the NG1000.

The modem interfaces between an external telephone line connection (via data line protector U8) and the system controller (A1). Its dc power source (+12 V) is provided by the the system controller (A1)

Figure 1.1: Block Diagram - NG1000 Transmitter System

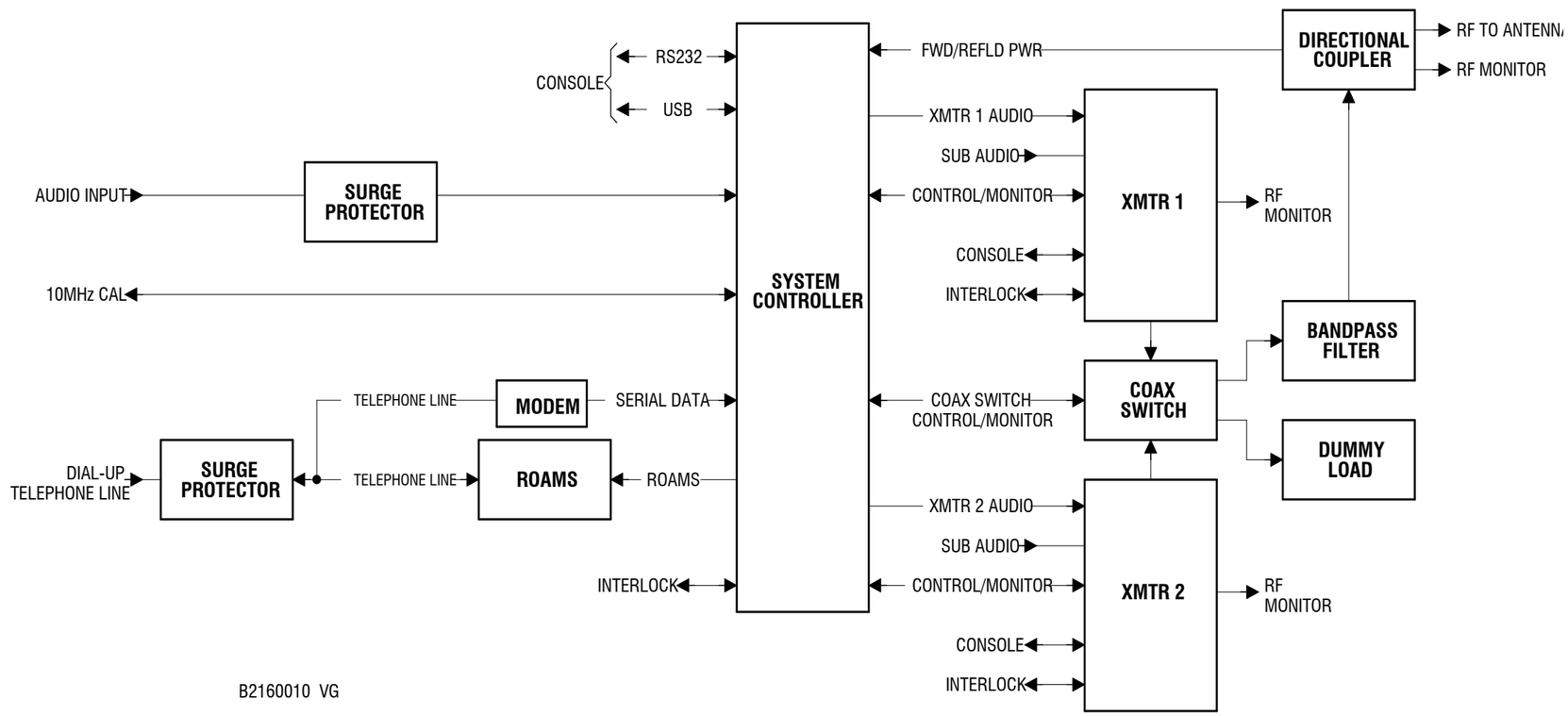
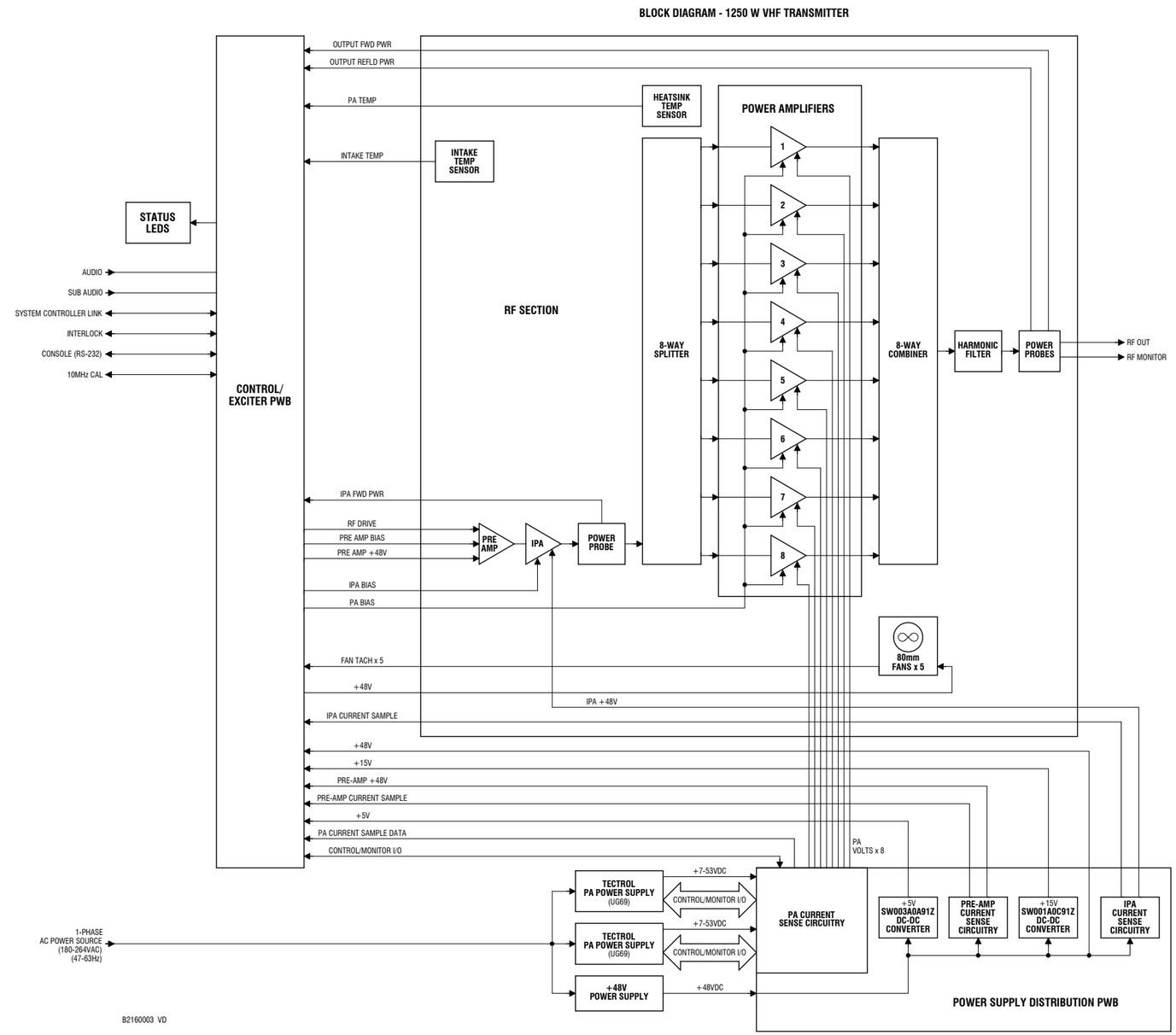


Figure 1.2: Block Diagram - 1250 W VHF Transmitter



Section 2: Operating the transmitter

This section provides information about operating the NG1000 transmitter:

- User interface - see page 2-2
  - Front panel UI - see page 2-2
  - NG series remote application - see page 2-47
- Top level menu - see page 2-8
- Main menu - see page 2-9
- Meters - see page 2-10
- Control - see page 2-13
  - Setting the active transmitter - see page 2-15
  - Setting system forward power - see page 2-15
  - Selecting channel frequency - see page 2-16
  - Turning on/off the standby transmitter - see page 2-16
  - Setting standby forward power - see page 2-17
  - Controlling VOX keying - see page 2-17
  - Setting changeover controls - see page 2-20
  - Low power tone control - see page 2-24
  - SAME control - see page 2-26
  - Setting FIPS code - see page 2-26
  - Setting time and date - see page 2-27
  - Trimming the carrier - see page 2-29
  - Setting the sub audio input gain - see page 2-29
  - Setting main audio parameters - see page 2-30
  - Setting temperature units - see page 2-31
  - Setting/resetting the system - see page 2-32
- Alarms - see page 2-33
- Events - see page 2-33
- Auxiliary I/O - see page 2-34
- Calibration - see page 2-38

User interface

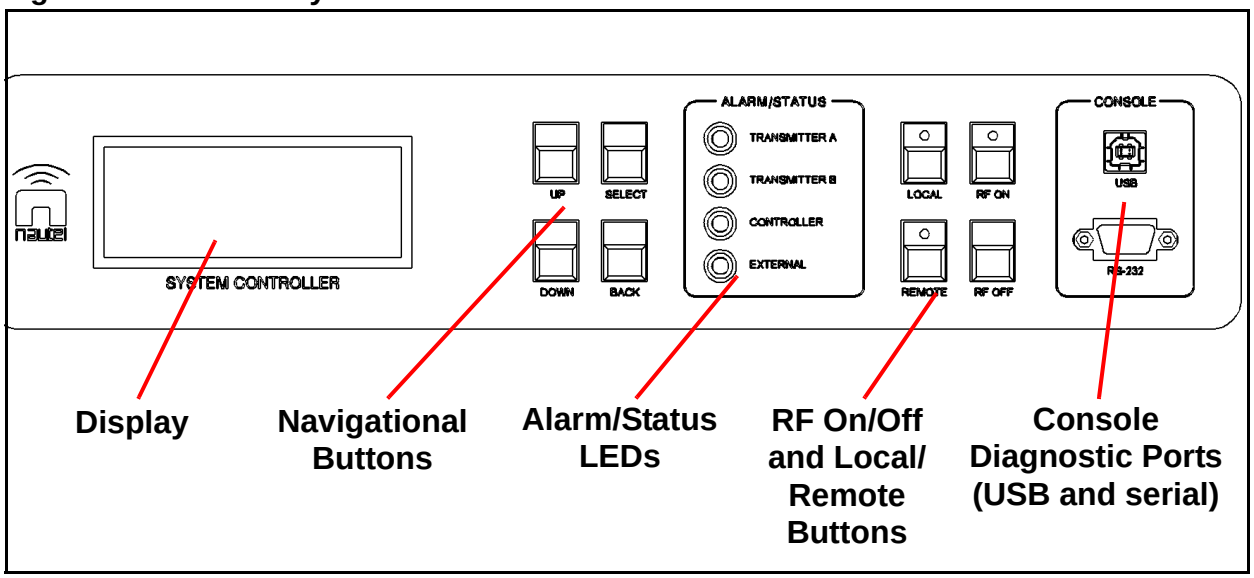
The user can interface with the NG1000 using one of two methods:

- Locally, using the front panel display and navigational buttons (see "Front panel UI").
- Remotely, using a laptop or PC that contains the "NG Remote Application" program (see "NG series remote application" on page 2-47)

Front panel UI

The NG1000 system's user interface (UI) consists primarily of items located on the front panel of the system controller unit (A1) (see Figure 2.1). There are also LEDs on the front panel of each VHF transmitter that indicate operational status. Similar interfacing is available on the remotely accessed "NG Remote Application" (see "NG series remote application" on page 2-47).

Figure 2.1: NG1000 System Controller Front Panel



LCD display

The primary interface is a 4 x 20 character vacuum fluorescent display (VFD). The display can be controlled by its adjacent pushbuttons. The display contains a menu-based series of pages that serve specific functions. See Figure 2.4 on page 2-5 for a flow diagram of these pages and Table 2.1 on page 2-5 for more detailed information.

Navigational buttons

There are four push-buttons on the right-hand side of the display that allow navigation through the display pages.

UP: moves cursor up through the associated page and increases the value of the associated selection on a parameter editing page

DOWN: moves cursor down through the associated page and decreases the value of the associated selection on a parameter editing page

SELECT: goes to the selected sub-menu screen, as applicable; acts as an "accept" or "save" function; display will prompt its use

BACK: returns to the next highest level screen; acts as a "cancel" function; display will prompt its use

Saving settings

Many of the display screens allow for modifying a selected setting. When you accept a setting change in an editing screen (using the SELECT button), the change is saved. To cancel a selection and return to the previous screen without saving, press the BACK button twice.

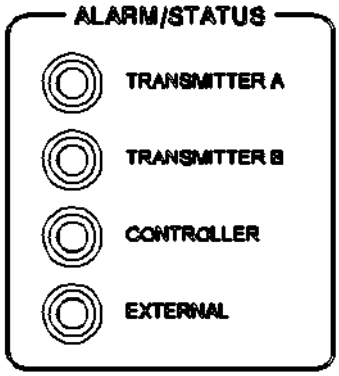
Alarm/status LEDs

There are four ALARM/STATUS LEDs on the system controller front panel - TRANSMITTER A, TRANSMITTER B, CONTROLLER and EXTERNAL (see Figure 2.2). These LEDs provide operational status of various sections of the transmitter system. The LEDs can glow green, red or blinking green/red.

For TRANSMITTER A or B LED, typically green indicates normal operation with RF on, red indicates RF off and blinking red/green indicates a fault or error.

For CONTROLLER or EXTERNAL LED, typically green indicates normal operation with RF on and red indicates a fault or error.

Figure 2.2: Alarm/Status LEDs



Refer to the Troubleshooting Manual for detailed information on LED status.

Local/remote buttons

Pressing the LOCAL push-button switch enables the transmitter system to be controlled via the system controller's front panel interface. All remote control circuits (except the external interlock) are disabled. When the transmitter system is in local mode, the LOCAL push-button switch's integral LED will be on.

Pressing the REMOTE push-button switch enables the transmitter system to be controlled via remote circuits. All push-button function switches on the system controller's front panel, except the RF OFF button, are disabled and have no influence. The RF OFF switch is still enabled to ensure the RF power stage can be turned off locally in the event of an emergency. You can still navigate the front panel display to view menus.

RF on/off buttons

The RF ON push-button switch turns on the active transmitter's RF power stage and cooling fans when pressed. Its integral lamp is on (green) when the RF power stage is on and the external interlock circuit is intact. The active transmitter will operate at its preset forward power setting.

The RF OFF push-button switch turns off the active and standby transmitter's RF power stages when pressed.

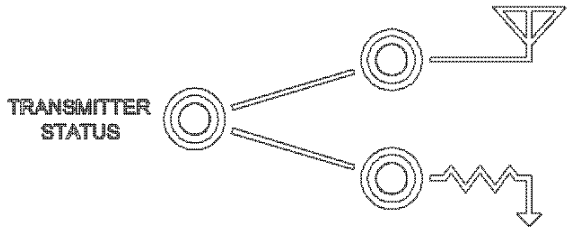
Console diagnostic ports

There are two CONSOLE ports on the front panel that provide options (USB and RS-232) to connect to a remote diagnostics console (see "NG series remote application" on page 2-47). These ports are also used to perform system software upgrades.

VHF transmitter status LEDs

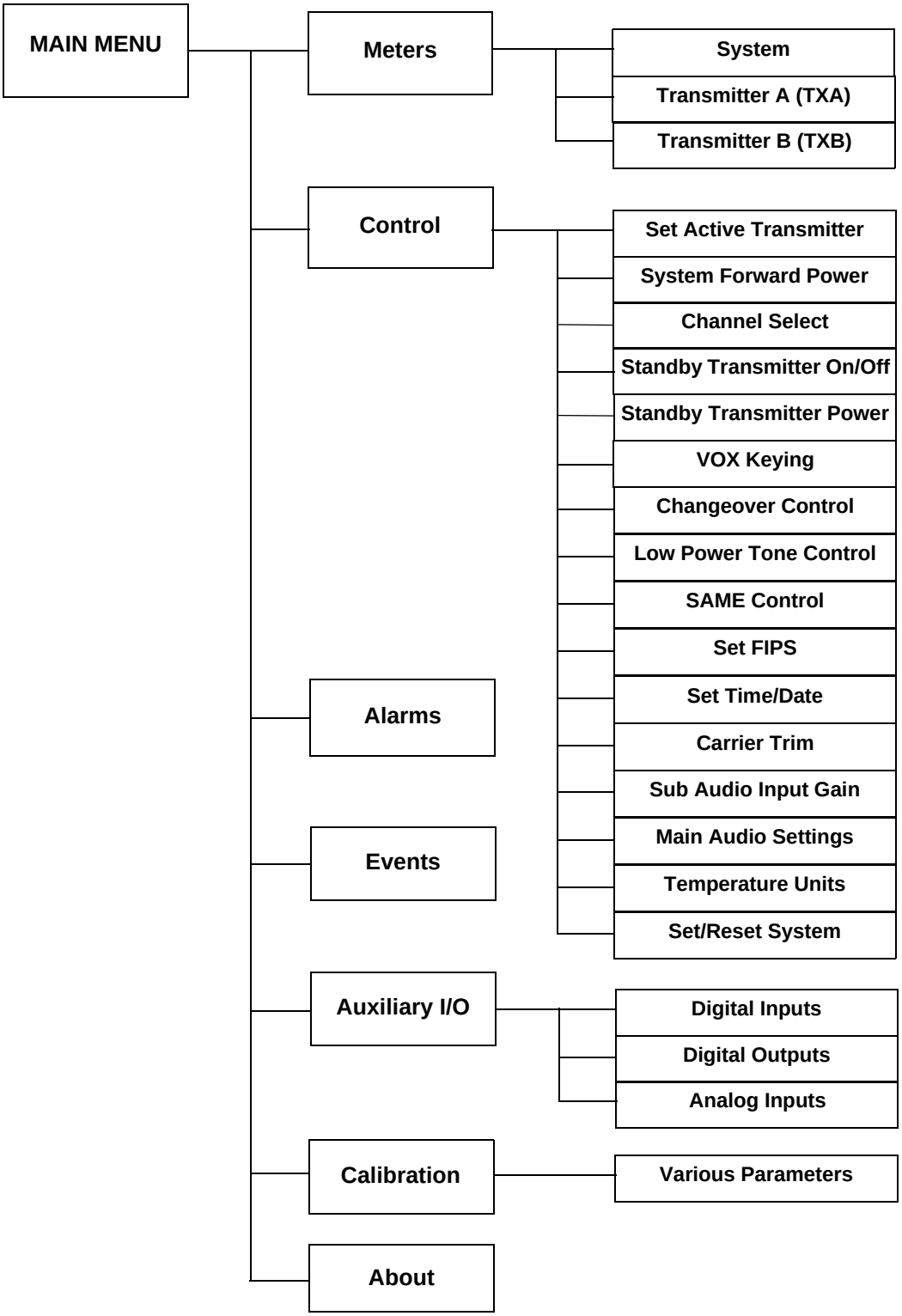
There are three LEDs on the front panel of each VHF transmitter - TRANSMITTER STATUS, antenna and load (see Figure 2.3). These LEDs provide operational and RF routing status for the associated transmitter. Typically, green indicates normal operation with RF on, red indicates RF off and blinking red/green indicates a fault or error. The status of the TRANSMITTER STATUS LED should be identical to the TRANSMITTER A or TRANSMITTER B LED on the system controller front panel.

Figure 2.3: Transmitter Status LEDs



Refer to the Troubleshooting Manual for detailed information on LED status.

Figure 2.4: NG1000 Flow Diagram



Display pages

Table 2.1 defines the available display pages.

Table 2.1: Display Pages

Display Page	Function	See Page
Top Level	View date/time, active transmitter and the transmitter system's forward and reflected power levels.	page 2-8
Main Menu	Allows entry to six sub-menus - Meters, Control, Alarms, Events, Calibration and About	page 2-9
Meters	Allows entry to three sub-menus - System, TXA (transmitter A) and TXB (transmitter B).	page 2-54
Meters - System	View the system's operational parameters including forward power, reflected power, measured frequency, cabinet temperature, low level dc supply voltages and input ac voltage.	page 2-11
Meters - TXA or TXB	View transmitter A or B's operational parameters.	page 2-12
Control	Allows entry to various sub-menus (see below).	page 2-13
Control - Set Active TX	Allows selection of the transmitter (A or B) that is routed to the system antenna.	page 2-15
Control - System Fwd Pwr	Allows setting of the active transmitter's forward power level.	page 2-15
Control - Channel Select	Allows setting of the active transmitter's operating carrier frequency.	page 2-16
Control - Standby TX On/ Off	Allows enabling and disabling of the standby transmitter's RF power stage.	page 2-16
Control - Standby TX Fwd Power	Allows setting of the standby transmitter's forward power level.	page 2-17
Control - VOX Keying	Allows control of the VOX keying feature.	page 2-17
Control - Changeover Control	Allows configuration of the system's main/standby transmitter changeover circuitry.	page 2-20

Table 2.1: Display Pages

Display Page	Function	See Page
Control - Low Power Tone Ctrl	Allows control of a low power tone that can be added to the modulation signal when the output power falls below a preset threshold.	page 2-24
Control - SAME Control	Allows enabling and disabling of the SAME (Specific Area Message Encoding) control feature.	page 2-26
Control - Set FIPS	Allows setting of the FIPS (Federal Information Processing Standards) code for your area.	page 2-26
Control - Set Time/Date	Set the system date and time.	page 2-27
Control - Carrier Trim	Allows adjustment of the system's carrier frequency.	page 2-29
Control - Sub Aud Input Lvl	Allows setting of the sub-audible input level.	page 2-29
Control - Main Audio Settings	Allows setting of the main audio input's modulation gain and enabling/disabling of the AGC feature.	page 2-30
Control - Temperature Units	Allows selection of Celsius or Fahrenheit as the system's displayed temperature unit.	page 2-31
Control - Set/Reset	Allows resetting (turn off, clear alarms) and setting (turn on) of the transmitter system.	page 2-32
Alarms	View the system's active alarms.	page 2-33
Events	View up to 254 most recent events (commands, alarms, etc.) that affected the transmitter system.	page 2-33
Auxiliary I/O	Allows configuration of user-determined digital inputs, digital outputs and analog inputs.	page 2-34
Calibration	Allows calibration of various system and transmitter parameters including forward power, reflected power, 10 MHz reference signals and pre-amp/IPA bias levels.	page 2-38
About	Displays the current software version installed in the system controller and the VHF transmitters' exciter/ control PWBs.	N/A

Top level menu

The display's top level menu is shown in Figure 2.5.

Figure 2.5: Top Level Menu

```
01/06/2010  08:00:00
Active TX: A
Fwd/Rfl: 0W/0W
SELECT = Main Menu
```

This menu displays the following information:

- Line 1 displays the current date and time.
- Line 2 displays the transmitter (A or B) that is operating into the antenna system.
- Line 3 displays the transmitter system's forward and reflected power levels.
- Line 4 indicates that pressing the SELECT button will enter the Main Menu (see Main menu on page 2-9).

Main menu

All navigation through the display menus begins on the MAIN MENU shown in Figure 2.6.

Figure 2.6: Main Menu

```
Main Menu:
>Meters
 Control
▼Alarms
```

This menu has seven sub-menu options:

- Meters - allows monitoring of NG1000 system status including operational parameters for the system and transmitters A and B (see Meters on page 2-10).
- Control - allows for various system controls including setting the active transmitter, setting forward power, setting frequency, enabling the standby transmitter, setting the time and date, setting/resetting the system (see Control on page 2-13).
- Alarms - allows monitoring of existing system alarms to aid in system troubleshooting (see Alarms on page 2-33).
- Events - provides a log that allows monitoring of current and past system events (alarms and commands) (see Events on page 2-33).
- Auxiliary I/O - allows configuration of digital inputs, digital outputs and analog outputs on the system controller's rear panel AUX I/O (A1J11A) connector (see Auxiliary I/O on page 2-34).
- Calibration - allows calibration of various system and transmitter parameters (see Calibration on page 2-38).
- About - displays the current version of software installed on the system controller and the transmitter exciter/control PWBs.

Use the UP and DOWN buttons to move the cursor (>) to the desired option and then press the SELECT button to open the selected sub-menu.

When a UI menu has more line items than can fit on the display, up and/or down directional arrows will appear to the left of the text to indicate which direction to scroll to view additional items.

Meters

You can monitor various system and transmitter parameters of the NG1000 in the Meters menu (see Figure 2.7).

Figure 2.7: Meters Menu

```
Meters:
>System
 TXA: Active
 TXB: Standby
```

This menu has three sub-menu options:

- System - allows monitoring of system operational parameters including forward power, reflected power, measured frequency, cabinet temperature, low level dc supply voltages and input ac voltage (see "System" on page 2-11).

- TXA - allows monitoring of transmitter A operational parameters including forward power, reflected power, PA voltages, current levels, operating temperatures and low level dc supply voltages (see "TXA (Transmitter A) or TXB (Transmitter B)" on page 2-12). The display also indicates whether transmitter A is operating as the system's Active or Standby transmitter.

- TXB - allows monitoring of transmitter B operational parameters including forward power, reflected power, PA voltages, current levels, operating temperatures and low level dc supply voltages (see "TXA (Transmitter A) or TXB (Transmitter B)" on page 2-12). The display also indicates whether transmitter B is operating as the system's Active or Standby transmitter.

Use the UP and DOWN buttons to move the cursor (>) to the desired option and then press the SELECT button to open the selected sub-menu.

When a UI menu has more line items than can fit on the display, up and/or down directional arrows will appear to the left of the text to indicate which direction to scroll to view additional items.

System

When you select System from the Meters menu (see Figure 2.8) you can view various system level parameters of the NG1000. These parameters are displayed on multiple pages of System screens.

Figure 2.8: Meters - System (Multiple Screens)

```
Meters: System:
 Fwd Pwr:0W
 Rfl Pwr:0W
▼VSWR:N/A
```

Use the UP and DOWN buttons to scroll through the following parameters in the System list:

- Fwd Pwr (forward power)
- Rfl Pwr (reflected power)
- VSWR
- Meas F (measured frequency)
- Cabinet Temp
- +5VDC
- +12VDC
- Primary VAC
- Secondary VAC
- AIn1 (analog input 1) through AIn4 (analog input 4)

TXA (Transmitter A) or TXB (Transmitter B)

When you select TXA or TXB from the Meters menu (see Figure 2.9) you can view various transmitter level parameters of the NG1000. These parameters are displayed on various pages of screens. Note that Active or Standby will also be displayed next to TXA or TXB to indicate which transmitter is operating into the system antenna.

Figure 2.9: Status - TXA or TXB Screen (Multiple Screens)

```
Meters: TXA: Active
 Fwd Pwr:0W
 Rfl Pwr:0W
▼VSWR:N/A
```

Use the UP and DOWN buttons to scroll through the following parameters listed on the TXA or TXB screens.

Fwd Pwr (forward power)	-15VDC
Rfl Pwr (reflected power)	+5VDC
VSWR	+3.3VDC
Preamp Curr (pre-amplifier current)	+1.8VDC
IPA Current	Fan Tach n (n=1-5)
IPA Fwd Pwr	Main Audio
Total PA Cur	Sub Audio
PA Voltsn (n=A, B)	Peak Dev
PA n Current (n=1-8)	Active PA Pwr
Intake Temp	PA DC-RF Eff
PA Heatsink (temp)	PA Diss Pwr
+48VDC	
+15VDC	

Control

You can control various operational functions of the NG1000 transmitter system in the Control menu (see Figure 2.10).

Figure 2.10: Control Menu (Multiple Screens)

```
Control:
>Set Active TX
 System Fwd Power
▼Channel Select
```

This menu has the following sub-menu options:

- Set Active TX - allows selection of the transmitter (A or B) that is routed to the system antenna (see "Setting the active transmitter" on page 2-15).
- System Fwd Power - allows setting of the active transmitter's forward power level (see "Setting system forward power" on page 2-15).
- Channel Select - allows setting of the active transmitter's operating frequency (see "Selecting channel frequency" on page 2-16).
- Standby TX On/Off - allows enabling and disabling of the standby transmitter's RF power stage (see "Turning on/off the standby transmitter" on page 2-16).
- Standby TX Fwd Power - allows setting of the standby transmitter's forward power level (see "Setting standby forward power" on page 2-17).
- VOX Keying - allows control of the VOX keying feature (see "Controlling VOX keying" on page 2-17).
- Changeover Ctrl - allows configuration of the system's main/standby transmitter changeover circuitry (see "Setting changeover controls" on page 2-20).
- Low Power Tone Ctrl - allows control of a low power tone that can be added to the modulation signal when the output power falls below a preset threshold (see "Low power tone control" on page 2-24).

- SAME Control - allows enabling and disabling of the SAME control feature (see "SAME control" on page 2-26).
- Set FIPS - allows setting of the FIPS code (see "Setting FIPS code" on page 2-26).
- Set Time/Date - allows setting the current time and date that is displayed on the first line of the top level menu (see "Setting time and date" on page 2-27).
- Carrier Trim - allows for minor adjustment of the carrier frequency (see "Trimming the carrier" on page 2-29).
- Sub Aud Input Gain - allows setting of the sub-audible input's gain (see "Setting the sub audio input gain" on page 2-29).
- Main Audio Settings - allows setting of the main audio's modulation gain percentage or enabling/disabling AGC (see "Setting main audio parameters" on page 2-30).
- Temperature Units - allows selection of Celsius or Fahrenheit as the system's displayed temperature unit (see "Setting temperature units" on page 2-31).
- Set/Reset System - allows for a set or reset of the transmitter system (see "Setting/resetting the system" on page 2-32).

Use the UP and DOWN buttons to move the cursor (>) to the desired option and then press the SELECT button to open the selected sub-menu.

When a UI menu has more line items than can fit on the display, up and/or down directional arrows will appear to the left of the text to indicate which direction to scroll to view additional items.

Setting the active transmitter

When you select Set Active TX from the Control menu (see Figure 2.11) you can determine the transmitter (A or B) that is routed to the system antenna as the active transmitter.

Figure 2.11: Set Active Transmitter Screen

```
Set Active TX:
>TXA
 TXB
  SELECT = Confirm
```

To change the active transmitter, use the UP and DOWN buttons to highlight the desired transmitter. Press the SELECT button to confirm the change. Press BACK to return to the Control menu.

Setting system forward power

When you select System Fwd Power from the Control menu (see Figure 2.12) you can set the desired forward power of the active transmitter that is being routed to the antenna.

Figure 2.12: System Forward Power Screens

```
System Fwd Power:
      1000 W
   (200 to 1000 W)
  SELECT = Confirm
```

```
System Fwd Power:*
       999 W
   (200 to 1000 W)
  SELECT = Confirm
```

To change the system forward power, use the UP and DOWN buttons to increase or decrease the power setting (between 200 W and 1000 W, in 1 W steps). Press and hold the UP or DOWN button to accelerate the rate of change. Note that an increase or decrease from the previously stored value will cause an asterisk to appear at the end of the screen's top line (see Figure 2.12). Press SELECT to confirm the setting. To cancel the setting and revert back to the previously stored value, press BACK, then press BACK again to return to the Control menu.

Selecting channel frequency

When you select Channel Select from the Control menu (see Figure 2.13) you can set the desired channel frequency of the active transmitter that is being routed to the antenna.

Figure 2.13: Channel Select Screens

```
Channel Select:
    162.425 MHz
(162.4 - 162.55 MHz)
  SELECT = Confirm
```

```
Channel Select:*
    162.450 MHz
(162.4 - 162.55 MHz)
  SELECT = Confirm
```

To change the channel frequency, use the UP and DOWN buttons to increase or decrease the frequency setting (between 162.4 and 162.55 MHz, in 25 kHz steps). Note that an increase or decrease from the previously stored value will cause an asterisk to appear at the end of the screen's top line (see Figure 2.13) Press SELECT to confirm the setting. To cancel the setting and revert back to the previously stored value, press BACK, then press BACK again to discard the changes and return to the previous menu.

> *NOTE:*
> *Unless another frequency was specified when ordered, the transmitter system's output cavity filter is factory tuned at 162.475 MHz. If a frequency change is required, perform the Frequency Change Procedure in Section 4, "Non-standard maintenance".*

Turning on/off the standby transmitter

When you select Stndby TX On/Off from the Control menu (see Figure 2.14) you can enable (turn on) or disable (turn off) the RF power stage of the transmitter routed to the system dummy load.

Figure 2.14: Standby TX On/Off Screen

```
Stndby TX On/Off:
>Off
 On
  SELECT = Confirm
```

To change the on/off selection, use the UP and DOWN buttons to highlight On or Off. Press the SELECT button to confirm the change.

Setting standby forward power

When you select Stndby TX Fwd Power from the Control menu (see Figure 2.15) you can set the desired forward power of the standby transmitter that is being routed to the system dummy load.

Figure 2.15: Standby Transmitter Forward Power Screen

```
Stndby TX Fwd Power:
       300 W
   (300 to 1250 W)
  SELECT = Confirm
```

To change the standby transmitter forward power, use the UP and DOWN buttons to increase or decrease the power setting (between 300 W and 1250 W, in 1 W steps). Press and hold the UP or DOWN button to accelerate the rate of change. Press SELECT to confirm the setting. To cancel the setting and revert back to the previously stored value, press BACK, then press BACK again to discard the changes and return to the previous menu.

Controlling VOX keying

The NG1000 transmitter system supports VOX keying. When you select VOX Keying from the Control menu (see Figure 2.16) you can control the enable/disable status, the keying timeout period and the keying level.

Figure 2.16: VOX Keying Menu

```
VOX Keying:
>Enable/Disable
 VOX Keying Timeout
 VOX Keying Level
```

This menu has three sub-menu options:

- Enable/Disable - allows enabling or disabling of VOX keying (see "Enabling/disabling VOX keying" on page 2-18).
- VOX Keying Timeout - allows setting of the VOX keying timeout period (see "Setting VOX keying timeout" on page 2-18).

- VOX Keying Level - allows setting of the VOX keying level (see "Setting VOX keying level" on page 2-19).

Use the UP and DOWN buttons to move the cursor to the desired option and then press the SELECT button to open the selected sub-menu.

Enabling/disabling VOX keying

When you select Enable/Disable VOX Keying from the VOX Keying menu (see Figure 2.17) you can enable (turn on) or disable (turn off) the use of VOX keying.

Figure 2.17: Enable/Disable VOX Keying Screen

```
VOX Keying:
>Disable
 Enable
  SELECT = Confirm
```

The factory default setting is Disable. To change the enable/disable selection, use the UP and DOWN buttons to highlight Disable or Enable. Press the SELECT button to confirm the change.

Setting VOX keying timeout

When you select VOX Keying Timeout from the VOX Keying menu (see Figure 2.18) you can set the timeout period for the VOX keying function.

Figure 2.18: VOX Keying Timeout Screens

```
VOX Keying Timeout:
      11 sec
   (10 to 60 sec)
  SELECT = Confirm
```

```
VOX Keying Timeout:*
      12 sec
   (10 to 60 sec)
  SELECT = Confirm
```

To change the VOX keying timeout period, use the UP and DOWN buttons to increase or decrease the timeout setting (between 10 and 60 seconds, in 1 second steps). Note that an increase or decrease from the previously stored value will cause an asterisk to appear at the end of the screen's top line (see Figure 2.18). Press SELECT to confirm the setting. To cancel the setting and revert back to the previously stored value, press BACK, then press BACK again to discard the changes and return to the previous menu.

Setting VOX keying level

When you select VOX Keying level from the VOX Keying menu (see Figure 2.19) you can set the level for the VOX keying function.

Figure 2.19: VOX Keying Level Screens

```
VOX Keying Level:
      -50 dBm
  (-50 to -25 dBm)
  SELECT = Confirm
```

```
VOX Keying Level:*
      -49 dBm
  (-50 to -25 dBm)
  SELECT = Confirm
```

To change the VOX keying level, use the UP and DOWN buttons to increase or decrease the level (between -50 and -25 dBm, in 1 dBm steps). Note that an increase or decrease from the previously stored value will cause an asterisk to appear at the end of the screen's top line (see Figure 2.19). Press SELECT to confirm the setting. To cancel the setting and revert back to the previously stored value, press BACK, then press BACK again to discard the changes and return to the previous menu.

Setting changeover controls

The NG1000 transmitter system supports a main-standby changeover feature for the two transmitters. When you select Changeover Ctrl from the Control menu (see Figure 2.20) you can control the enable/disable status for two types of changeover and establish the threshold levels at which the changeovers should occur.

Figure 2.20: Changeover Control Menu

```
Changeover Ctrl:
>Auto Changeover
 Tone Changeover
```

This menu has two sub-menu options:

- Auto Changeover - allows enabling or disabling of automatic changeovers, setting the forward power threshold and the changeover delay (see "Setting up changeover parameters").
- Tone Changeover - allows enabling or disabling of a tone changeover option, as well as a means to set the tone's changeover level (see "Changeover Delay Screen" on page 2-22).

Use the UP and DOWN buttons to move the cursor to the desired option and then press the SELECT button to open the selected sub-menu.

Setting up changeover parameters

When you select Auto Changeover from the Changeover Ctrl menu (see Figure 2.25) you enable or disable a feature that allows the system to cause a transmitter changeover if the system output power falls below the Fwd Pwr (forward power) Threshold.

Figure 2.21: Auto Changeover Screen

```
Auto Changeover:
>Enable/Disable
 Fwd Pwr Threshold
 Changeover Delay
```

This menu has three sub-menu options:

- Enable/Disable - allows enabling or disabling of tone changeovers (see Figure 2.22).
- Fwd Pwr Threshold - allows setting of the forward power threshold below which a changeover will occur (see "Setting forward power threshold").
- Changeover Delay - allows setting of the time period that the forward power is below the changeover threshold before a changeover will occur (see Figure 2.24 on page 2-22).

Use the UP and DOWN buttons to move the cursor to the desired option and then press the SELECT button to open the selected sub-menu.

Figure 2.22: Auto Changeover Enable Screen

```
Auto Changeover:
>Disable
 Enable
  SELECT = Confirm
```

The factory default setting is Enable. To change the enable/disable selection, use the UP and DOWN buttons to highlight Disable or Enable. Press the SELECT button to confirm the change.

Setting forward power threshold

When you select Fwd Pwr Threshold from the Auto Changeover menu (see Figure 2.23) you can set the forward power level at which an automatic changeover - if enabled - will occur. When the active transmitter's forward power level falls below the threshold, the system controller will initiate a changeover to the standby transmitter.

Figure 2.23: Forward Power Threshold Screens

```
Fwd Pwr Threshold:
        90 %
    (50 to 90 %)
  SELECT = Confirm
```

```
Fwd Pwr Threshold:*
        89 %
    (50 to 90 %)
  SELECT = Confirm
```

To change the forward power threshold, use the UP and DOWN buttons to increase or decrease the threshold (between 50 and 90%, in 1 % steps). The factory default setting is 90%. Note that an increase or decrease from the previously stored value will cause an asterisk to appear at the end of the screen's top line (see Figure 2.23). Press SELECT to confirm the setting. To cancel the setting and revert back to the previously stored value, press BACK, then press BACK again to discard the changes and return to the previous menu.

Setting changeover delay

When you select Changeover Delay from the Auto Changeover menu (see Figure 2.24) you can set the changeover delay after which an automatic changeover - if enabled - will occur. When the active transmitter's forward power level falls below the forward power threshold for a period of time equal to the changeover delay, the system controller will initiate a changeover to the standby transmitter.

Figure 2.24: Changeover Delay Screen

```
Changeover Delay:
        5 s
    (2 to 20 s)
  SELECT = Confirm
```

The factory default setting is 5 s. To change the changeover delay, use the UP and DOWN buttons to change the delay period to any value between 2 seconds and 20 seconds. Press the SELECT button to confirm the change.

Setting up for tone changeover

When you select Tone Changeover from the Changeover Ctrl menu (see Figure 2.25) you can enable or disable a feature that allows a user-supplied tone to cause a transmitter changeover. This tone is applied within the audio stream and its trigger threshold is also established within the Tone Changeover menu.

Figure 2.25: Tone Changeover Screen

```
Tone Changeover:
>Enable/Disable
 Chgovr Tone Level
```

This menu has two sub-menu options:

- Enable/Disable - allows enabling or disabling of tone changeovers (see Figure 2.26).
- Chgovr Tone Level - allows setting of the tone level at which a changeover will occur (see "Setting changeover tone level").

Use the UP and DOWN buttons to move the cursor to the desired option and then press the SELECT button to open the selected sub-menu.

Figure 2.26: Tone Changeover Enable/Disable Screen

```
Tone Changeover:
>Disable
 Enable
  SELECT = Confirm
```

The factory default setting is Enable. To change the enable/disable selection, use the UP and DOWN buttons to highlight Disable or Enable. Press the SELECT button to confirm the change.

Setting changeover tone level

When you select Chgovr Tone level from the Tone Changeover menu (see Figure 2.27 on page 2-23) you can set the level at which a tone will cause a changeover.

Figure 2.27: Tone Changeover Level Screens

```
Chgovr Tone Level:
       0 dBm
   (-20 to 0 dBm)
  SELECT = Confirm
```

```
Chgovr Tone Level:*
       -1 dBm
   (-20 to 0 dBm)
  SELECT = Confirm
```

To change the changeover tone level, use the UP and DOWN buttons to increase or decrease the level (between -20 and 0 dBm, in 1 dBm steps). Note that an increase or decrease from the previously stored value will cause an asterisk to appear at the end of the screen's top line (see Figure 2.27). Press SELECT to confirm the setting. To cancel the setting and revert back to the previously stored value, press BACK, then press BACK again to discard the changes and return to the previous menu.

Low power tone control

When you select Low Power Tone Ctrl from the Control menu (see Figure 2.28) you can set the low power tone level and threshold. When the output power cannot be maintained above the threshold, a 425 Hz tone is added to the modulation signal. The level is referenced to 100% modulation, and is nominally set to -16 dBFS (15.8% modulation).

Figure 2.28: Low Power Tone Control Menu

```
Low Power Tone Ctrl:
>Low Pwr Tone Dev
 Low Pwr Tone Thresh
```

This menu has two sub-menu options:

- Low Pwr Tone Dev - allows setting the acceptable frequency deviation of the low power tone (see "Setting low power tone deviation" on page 2-25).
- Low Pwr Tone Thresh - allows setting the threshold for the low power tone (see "Setting low power tone threshold" on page 2-25).

Use the UP and DOWN buttons to move the cursor to the desired option and then press the SELECT button to open the selected sub-menu.

Setting low power tone deviation

When you select Low Pwr Tone Dev from the Low Pwr Tone Ctrl menu (see Figure 2.29) you can set the frequency deviation for a low power tone.

Figure 2.29: Low Power Tone Level Screens

```
Low Pwr Tone Dev:
      792 Hz
   (0 to 5000 Hz)
 SELECT = Confirm
```

```
Low Pwr Tone Dev:*
      793 Hz
   (0 to 5000 Hz)
 SELECT = Confirm
```

To change the low power tone deviation, use the UP and DOWN buttons to increase or decrease the frequency (between 0 and 5000 Hz, in 1 Hz steps). Note that an increase or decrease from the previously stored value will cause an asterisk to appear at the end of the screen's top line (see Figure 2.29). Press SELECT to confirm the setting. To cancel the setting and revert back to the previously stored value, press BACK, then press BACK again to discard the changes and return to the previous menu.

Setting low power tone threshold

When you select Low Pwr Tone Thresh from the Low Pwr Tone Ctrl menu (see Figure 2.30) you can set the threshold for the low power tone. When the output power of the system cannot be maintained above this level, the low power tone will be added to the modulating signal.

Figure 2.30: Low Power Tone Threshold Screens

```
Low Pwr Tone Thresh:
       60 %
   (25 to 75 %)
 SELECT = Confirm
```

```
Low Pwr Tone Thresh:*
       61 %
   (25 to 75 %)
 SELECT = Confirm
```

To change the low power tone threshold, use the UP and DOWN buttons to increase or decrease the threshold (between 25 and 75%, in 1 % steps). Note that an increase or decrease from the previously stored value will cause an asterisk to appear at the end of the screen's top line (see Figure 2.30). Press SELECT to confirm the setting. To cancel the setting and revert back to the previously stored value, press BACK, then press BACK again to discard the changes and return to the previous menu.

SAME control

The NG1000 transmitter system supports the use of SAME (Specific Area Message Encoding) technology. With this technology, compatible receivers can localize (i.e., filter out non-local information) weather alerts to specific areas. When you select SAME Control from the Control menu (see Figure 2.31), you can enable or disable this feature.

Figure 2.31: SAME Control Menu

```
SAME Control:
>Disable
 Enable
  SELECT = Confirm
```

To change the enable/disable selection, use the UP and DOWN buttons to highlight Disable or Enable. Press the SELECT button to confirm the change.

Setting FIPS code

When you select Set FIPS from the Control menu (see Figure 2.32) you can set the FIPS (Federal Information Processing Standards) code for your designated area.

Figure 2.32: Set FIPS Screens

```
Set FIPS:
       010000

  SELECT = Confirm
```

```
Set FIPS:*
       009999

  SELECT = Confirm
```

To change the FIPS code, use the UP and DOWN buttons to increase or decrease the value as necessary. Press and hold the UP or DOWN button to accelerate the rate of change. Note that an increase or decrease from the previously stored value will cause an asterisk to appear at the end of the screen's top line (see Figure 2.32). Press SELECT to confirm the setting. To cancel the setting and revert back to the previously stored value, press BACK, then press BACK again to discard the changes and return to the previous menu.

Setting time and date

When you select Set Time/Date from the Control menu (see Figure 2.33) you can set the current time and date. The time and date appears on the top line of the top level menu (see Figure 2.5 on page 2-8).

Figure 2.33: Set Time/Date Screen

```
Set Time/Date:
>Set Time 13:40:16
 Set Date 01/06/2010
```

This menu has two sub-menu options:

- Set Time - allows setting the current time (see "Setting time").
- Set Date - allows setting the current date (see "Setting date" on page 2-28).

Setting time

When you select Set Time from the Set Time/Date menu (see Figure 2.34) you can set the current time for the transmitter system.

Figure 2.34: Set Time Screens

```
Set Time:
 <13>: 40 : 16
   SELECT = Next
   BACK = Cancel
```

```
Set Time:
  13 :<40>: 16
   SELECT = Next
  BACK = Previous
```

```
Set Time:
  13 : 40 :<16>
  SELECT = Confirm
  BACK = Previous
```

To change the time, use the SELECT button to move between the time fields (hours : minutes : seconds), and use the UP and DOWN buttons to increase or decrease the value of the highlighted field as desired. After editing the seconds field, press SELECT to confirm the time setting. To cancel the setting and revert back to the previously stored value, press BACK through the time fields to discard the changes and return to the previous menu.

Setting date

When you select Set Date from the Set Time/Date menu (see Figure 2.35) you can set the current date for the transmitter system.

Figure 2.35: Set Date Screens

```
Set Date:
 <01>/ 06 / 2010
   SELECT = Next
   BACK = Cancel
```

```
Set Date:
  01 /<06>/ 2010
   SELECT = Next
  BACK = Previous
```

```
Set Date:
  01 / 06 /<2010>
  SELECT = Confirm
  BACK = Previous
```

To change the date, use the SELECT button to move between the date fields (month / day / year), and use the UP and DOWN buttons to increase or decrease the value of the highlighted field as desired. After editing the year field, press SELECT to confirm the date setting. To cancel the setting and revert back to the previously stored value, press BACK through the date fields to discard the changes and return to the previous menu.

Trimming the carrier

When you select Carrier Trim from the Control menu (see Figure 2.36) you can trim the frequency of the transmitter system's carrier frequency.

Figure 2.36: Carrier Trim Screens

```
Carrier Trim:
      -2 Hz
 (-1000 to 1000 Hz)
  SELECT = Confirm
```

```
Carrier Trim:*
      -1 Hz
 (-1000 to 1000 Hz)
  SELECT = Confirm
```

To change the carrier trim value, use the UP and DOWN buttons to increase or decrease the carrier frequency as desired (between -1000 and 1000 Hz, in 1 Hz steps). Note that an increase or decrease from the previously stored value will cause an asterisk to appear at the end of the screen's top line (see Figure 2.36). Press SELECT to confirm the setting. To cancel the setting and revert back to the previously stored value, press BACK, then press BACK again to discard the changes and return to the previous menu.

Setting the sub audio input gain

When you select Sub Audio Input Gain from the Control menu (see Figure 2.37) you can set the gain of the sub-audible input. Typically, a 1 dBm tone will result in a 20% modulation level.

Figure 2.37: Sub Audio Input Gain Screens

```
Sub Aud Input Gain:
      4.0 dB
    (-6 to 6 dB)
  SELECT = Confirm
```

```
Sub Aud Input Gain:*
      5.0 dB
    (-6 to 6 dB)
  SELECT = Confirm
```

To change the sub-audible input gain, use the UP and DOWN buttons to increase or decrease the amplitude as desired (between -6 and 6 dBm, in 0.1 dBm steps). Note that an increase or decrease from the previously stored value will cause an asterisk to appear at the end of the screen's top line (see Figure 2.37). Press SELECT to confirm the setting. To cancel the setting and revert back to the previously stored value, press BACK, then press BACK again to discard the changes and return to the previous menu.

Setting main audio parameters

When you select Main Audio Settings from the Control menu (see Figure 2.38) you can set the main audio's modulation gain or enable/disable the automatic gain control (AGC) feature.

Figure 2.38: Main Audio Settings Screen

```
Main Audio Settings:
>Main Aud Mod Gain
 AGC Enable/Disable
```

This menu has two sub-menu options:

- Main Aud Mod Gain - allows setting the gain of the main audio's modulation. The value selected determines the peak modulation depth of the main audio (see "Setting the main audio modulation gain" on page 2-30).
- AGC Enable/Disable - allows enabling or disabling of the AGC feature (see "Enabling/disabling AGC" on page 2-31).

Setting the main audio modulation gain

See Figure 2.39.

Figure 2.39: Main Audio Mod Gain Screens

```
Main Aud Mod Gain:
      14 %
   (10 to 100 %)
 SELECT = Confirm
```

```
Main Aud Mod Gain:*
      15 %
   (10 to 100 %)
 SELECT = Confirm
```

To change the main audio modulation gain, use the UP and DOWN buttons to increase or decrease the percentage as desired (between 10 and 100 %, in 1 % steps). Note that an increase or decrease from the previously stored value will cause an asterisk to appear at the end of the screen's top line (see Figure 2.39). Press SELECT to confirm the setting. To cancel the setting and revert back to the previously stored value, press BACK, then press BACK again to discard the changes and return to the previous menu.

Enabling/disabling AGC

See Figure 2.40.

Figure 2.40: AGC Enable/Disable Screen

```
AGC Enable/Disable:
>Disable
 Enable
  SELECT = Confirm
```

The factory default setting is Disable. To change the enable/disable selection, use the UP and DOWN buttons to highlight Disable or Enable. Press the SELECT button to confirm the change.

Setting temperature units

When you select Temperature Units from the Control menu (see Figure 2.41), you can edit the displayed temperature unit of measure (Celsius or Fahrenheit).

Figure 2.41: Temperature Units Screen

```
Temperature Units:
>Celsius
 Fahrenheit
  SELECT = Confirm
```

The factory default setting is Celsius. To change the temperature units selection, use the UP and DOWN buttons to highlight Celsius or Fahrenheit. Press the SELECT button to confirm the change.

Setting/resetting the system

When you select Set/Reset System from the Control menu (see Figure 2.42), you can turn the system on and off and clear alarms.

Figure 2.42: Set/Reset System Screen

```
Set/Reset System:
>Set
 Reset
  SELECT = Confirm
```

Use the UP and DOWN buttons to highlight Set or Reset. Press the SELECT button to confirm the change.

Selecting Reset turns off the transmitter system and attempts to clear any alarms.

Selecting Set turns on the transmitter system.

Alarms

You can view all active transmitter system alarms using the Alarms screen(s) (see Figure 2.43).

Figure 2.43: Alarms Screen(s)

```
Alarms:          1-3/4
>Coax Switch Fault
 Coax Interlock Open
▼TXA Comm Fail
```

```
Alarms:          2-4/4
▲Coax Interlock Open
 TXA Comm Fail
 TXB Comm Fail
```

Use the UP and DOWN buttons to scroll through the list of active alarms, noting there may be multiple pages.

Refer to the *NG1000 Troubleshooting Manual* to cross-reference the alarm name to possible causes and troubleshooting tips.

Events

The most recent 254 events (alarms, commands, etc.) that affected the transmitter system are stored in memory. You can view these events using the Events screen(s) (see Figure 2.44).

Figure 2.44: Events Screen(s)

```
Events: 1 of 254
 01/06/2010 12:22:32
 Cmd LOCAL
▼ SELECT = Clear all
```

Use the UP and DOWN buttons to scroll through the list of events, noting there may be multiple pages. A list of events (1 is the most recent) will be displayed, sorted by date and time, with a description of the event.

Press the SELECT button to clear all displayed events from memory.

Auxiliary I/O

You can configure the logic for auxiliary remote inputs and outputs for the NG1000 transmitter system in the Auxiliary I/O menu (see Figure 2.45). The auxiliary inputs and outputs are connected to the AUX I/O (A1J11A) connector on the rear of the system controller..

Figure 2.45: Auxiliary I/O Menu

```
Auxiliary I/O:
>Digital Inputs
 Digital Outputs
 Analog Inputs
```

This menu has three sub-menu options:

- Digital Inputs - allows configuration of up to four digital inputs (see "Digital inputs" on page 2-35).
- Digital Outputs - allows configuration of up to four digital outputs (see "Digital outputs" on page 2-36).
- Analog Inputs - allows configuration of up to four analog inputs (see "Analog inputs" on page 2-36).

Use the UP and DOWN buttons to move the cursor to the desired option and then press the SELECT button to open the selected sub-menu.

Digital inputs

Using the Digital Inputs menu (see Figure 2.46), you can configure up to four remotely supplied digital inputs. These inputs must be connected by the user. They are for auxiliary use only and have no effect on transmitter operation.

Figure 2.46: Digital Inputs Screens

```
Digital Inputs:
>Dig In 1 (J11A-1)
 Dig In 2 (J11A-2)
▼Dig In 3 (J11A-3)
```

SELECT AN INPUT TO VIEW OR EDIT

```
Dig In 1 (J11A-1)
 Label:DIn1(J11A-1)
 Current state:High
>Alarm Sense:[ Low]
```

VIEW / EDIT SCREEN
Label and Current State are for viewing only

Use the UP and DOWN buttons to scroll through the digital inputs (Dig In 1 through Dig In 4) and press the SELECT button to open the selected input's view/edit screen (see Figure 2.46). The input name also includes its associated pin of the AUX I/O (A1J11A) connector.

In the digital input's view/edit screen, you can view the Label [e.g., DIn1 (J11A-1)], the current logic state (high or low) of the digital input and the alarm sensing logic (e.g., Low indicates that the digital input must be logic low to generate an alarm). Press the SELECT button to toggle the Alarm Sense state between High and Low. Press BACK to return to the previous menu.

If a digital input is in an alarm state, the front panel's ALARM/STATUS - EXTERNAL LED will turn red and the Alarms screen will display an alarm.

Digital input labels can be edited using the NG Series Remote Application (see "NG series remote application" on page 2-47). If a label is changed, it will also appear on the front panel UI.

Digital outputs

Using the Digital Outputs menu (see Figure 2.47), you can configure up to four digital outputs to provide either an open circuit or a ground potential. These outputs must be connected by the user. They are for auxiliary use only and are not representative of transmitter operation

Figure 2.47: Digital Outputs Screens

```
Digital Outputs:
>Dig Out 1 (J11A-7)
 Dig Out 2 (J11A-8)
▼Dig Out 3 (J11A-9)
```

SELECT AN OUTPUT TO VIEW OR EDIT

```
Dig Out 1 (J11A-7)
 Label:DOu1(J11A-7)
>Setting:[Open]
```

VIEW / EDIT SCREEN
Label is for viewing only

Use the UP and DOWN buttons to scroll through the digital outputs (Dig Out 1 through Dig Out 4) and press the SELECT button to open the selected output's view/edit screen (see Figure 2.47). The output name also includes its associated pin of the AUX I/O (A1J11A) connector.

In the digital output's view/edit screen, you can view the Label [e.g., DOu1 (J11A-7)] and the setting (Open or GND) for the digital output (e.g., Open indicates that the digital output provides an open circuit; GND indicates that the digital output provides a ground potential). Press the SELECT button to toggle the Setting state between Open and GND. Press BACK to return to the previous menu.

Digital output labels can be edited using the NG Series Remote Application (see "NG series remote application" on page 2-47). If a label is changed, it will also appear on the front panel UI.

Analog inputs

Using the Analog Inputs menu (see Figure 2.48), you can configure up to four analog inputs. These inputs must be connected by the user. They are for auxiliary use only and have no effect on transmitter operation.

Figure 2.48: Analog Inputs Screens

```
Analog Inputs:
>Analog 1 (J11A-11)
 Analog 2 (J11A-12)
▼Analog 3 (J11A-13)
```

SELECT AN INPUT TO VIEW OR EDIT

```
Analog 1 (J11A-11)
 Label:AIn1(J11A-11)
>Threshold:   5.0V
 Alarm Sense: >
```

VIEW / EDIT SCREEN
Label is for viewing only

```
Analog 1 (J11A-11)
 Threshold
       5.0 V
  SELECT = Confirm
```

```
Analog 1 (J11A-11)
 Alarm Sense
>Input > Threshold
 Input < Threshold
```

Use the UP and DOWN buttons to scroll through the analog inputs (Analog 1 through Analog 4) and press the SELECT button to open the selected input's view/edit screen (see Figure 2.48). The input name also includes its associated pin of the AUX I/O (A1J11A) connector.

In the analog input's view/edit screen, you can view the Label [e.g., AIn1 (J11A-11)], the alarm threshold (user determined) and the alarm sensing logic (e.g., > indicates that the analog input must be greater than the preset threshold value to generate an alarm). Use the UP and DOWN buttons to place the cursor on Threshold or Alarm Sense and press the SELECT button.

In the Threshold screen, use the UP and DOWN buttons to adjust the analog input's alarm threshold. (between 0.0 V and 12.0 V). Press the SELECT button to confirm the threshold setting.

In the Alarm Sense screen, use the UP and DOWN buttons to select Input > Threshold or Input < Threshold. Press the SELECT button to confirm the alarm sense setting. Press BACK to return to the previous menu.

Analog input labels can be edited using the NG Series Remote Application (see "NG series remote application" on page 2-47). If a label is changed, it will also appear on the front panel UI.

Calibration

You can calibrate various parameters of the NG1000 transmitter system in the Calibration menu (see Figure 2.10).

Figure 2.49: Calibration Menu (Multiple Screens)

```
Calibration:
>System Fwd Pwr Cal
 System Rfl Pwr Cal
▼Sys Con 10MHz Cal
```

The following parameters can be calibrated using this menu:

- System Fwd Pwr Cal - see page 2-39
- System Rfl Pwr Cal - see page 2-39
- Sys Con 10MHz Cal - see page 2-40
- TXA Fwd Pwr Cal - see page 2-41
- TXA Rfl Pwr Cal - see page 2-41
- TXA Preamp/IPA Bias - see page 2-42
- TXA 10MHz Cal - see page 2-42
- TXA 10MHz Cal - see page 2-42
- TXA 10MHz Cal - see page 2-42
- TXA Audio input - see page 2-43
- TXA Subaudible input - see page 2-43
- TXB Fwd Pwr Cal - see page 2-44
- TXB Rfl Pwr Cal - see page 2-44
- TXB Preamp/IPA Bias - see page 2-45
- TXB 10MHz Cal - see page 2-45
- TXB 10MHz Cal - see page 2-45
- TXB 10MHz Cal - see page 2-45
- TXB Audio input - see page 2-46
- TXB Subaudible input - see page 2-46

NOTE: Parameters prefixed by Sys Con are part of the system controller; parameters prefixed by TXA or TXB are part of transmitter A or transmitter B.

System Fwd Pwr Cal

When you select System Fwd Pwr Cal from the Calibration menu (see Figure 2.50) you can calibrate the system's forward power level.

Figure 2.50: System Fwd Pwr Cal Screens

```
System Fwd Pwr Cal:
       1.000
 Fwd Pwr:0W
  SELECT = Confirm
```

```
System Fwd Pwr Cal:*
       1.001
 Fwd Pwr:0W
  SELECT = Confirm
```

To calibrate the system forward power, use the UP and DOWN buttons to increase or decrease the scale factor as desired. Note that an increase or decrease from the previously stored value will cause an asterisk to appear at the end of the screen's top line (see Figure 2.50). Press SELECT to confirm the setting. To cancel the setting and revert back to the previously stored value, press BACK, then press BACK again to discard the changes and return to the previous menu. Note that the forward power (Fwd Pwr) displayed on the screen does not change until you press SELECT to confirm the setting. You can then re-adjust the scale factor, as necessary.

System Rfl Pwr Cal

When you select System Rfl Pwr Cal from the Calibration menu (see Figure 2.51) you can calibrate the system's reflected power level.

Figure 2.51: System Rfl Pwr Cal Screens

```
System Rfl Pwr Cal:
       1.000
 Rfl Pwr:0W
  SELECT = Confirm
```

```
System Rfl Pwr Cal:*
       1.001
 Rfl Pwr:0W
  SELECT = Confirm
```

To calibrate the system reflected power, use the UP and DOWN buttons to increase or decrease the scale factor as desired. Note that an increase or decrease from the previously stored value will cause an asterisk to appear at the end of the screen's top line (see Figure 2.51). Press SELECT to confirm the setting. To cancel the setting and revert back to the previously stored value, press BACK, then press BACK again to discard the changes and return to the previous menu. Note that the reflected power (Rfl Pwr) displayed on the screen does not change until you press SELECT to confirm the setting. You can then re-adjust the scale factor, as necessary.

Sys Con 10MHz Cal

When you select Sys Con 10MHz Cal from the Calibration menu (see Figure 2.52) you can calibrate the system's 10 MHz reference.

Figure 2.52: Sys Con 10MHz Cal Screens

```
Sys Con 10MHz Cal:
        1
    (0 to 65535)
  SELECT = Confirm
```

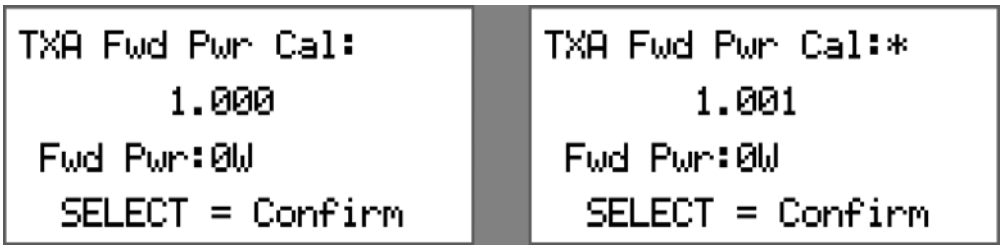
```
Sys Con 10MHz Cal:*
        2
    (0 to 65535)
  SELECT = Confirm
```

To calibrate the system 10 MHz signal, use the UP and DOWN buttons to increase or decrease the value as desired (between 0 and 65535). Press and hold the UP or DOWN button to accelerate the rate of change. Note that an increase or decrease from the previously stored value will cause an asterisk to appear at the end of the screen's top line (see Figure 2.52). Press SELECT to confirm the setting. To cancel the setting and revert back to the previously stored value, press BACK, then press BACK again to discard the changes and return to the previous menu.

TXA Fwd Pwr Cal

When you select TXA Fwd Pwr Cal from the Calibration menu (see Figure 2.53) you can calibrate transmitter A's forward power level.

Figure 2.53: TXA Fwd Pwr Cal Screens

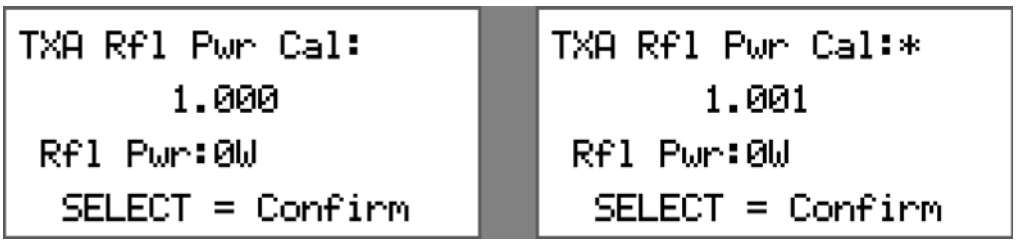


To calibrate transmitter A's forward power, use the UP and DOWN buttons to increase or decrease the scale factor as desired. Note that an increase or decrease from the previously stored value will cause an asterisk to appear at the end of the screen's top line (see Figure 2.53). Press SELECT to confirm the setting. To cancel the setting and revert back to the previously stored value, press BACK, then press BACK again to discard the changes and return to the previous menu. Note that the forward power (Fwd Pwr) displayed on the screen does not change until you press SELECT to confirm the setting. You can then re-adjust the scale factor, as necessary.

TXA Rfl Pwr Cal

When you select TXA Rfl Pwr Cal from the Calibration menu (see Figure 2.54) you can calibrate transmitter A's reflected power level.

Figure 2.54: TXA Rfl Pwr Cal Screens



To calibrate transmitter A's reflected power, use the UP and DOWN buttons to increase or decrease the scale factor as desired. Note that an increase or decrease from the previously stored value will cause an asterisk to appear at the end of the screen's top line (see Figure 2.54). Press SELECT to confirm the setting. To cancel the setting and revert back to the previously stored value, press BACK, then press BACK again to discard the changes and return to the previous menu. Note that the reflected power (Rfl Pwr) displayed on the screen does not change until you press SELECT to confirm the setting. You can then re-adjust the scale factor, as necessary.

TXA Preamp/IPA Bias

When you select TXA Preamp/IPA Bias from the Calibration menu (see Figure 2.55) you can calibrate transmitter A's pre-amplifier/IPA bias value.

Figure 2.55: TXA Preamp/IPA Bias Screen

```
TXA Preamp/IPA Bias

  Press SELECT to
 start the routine.
```

To calibrate transmitter A's pre-amplifier/IPA bias value, press the SELECT button to start a routine that will automatically callibrate the bias value. When the routine is complete (i.e., progress is 100%), press BACK to return to the previous menu.

TXA 10MHz Cal

When you select TXA 10MHz Cal from the Calibration menu (see Figure 2.56) you can calibrate transmitter A's 10 MHz reference.

Figure 2.56: TXA 10MHz Cal Screens

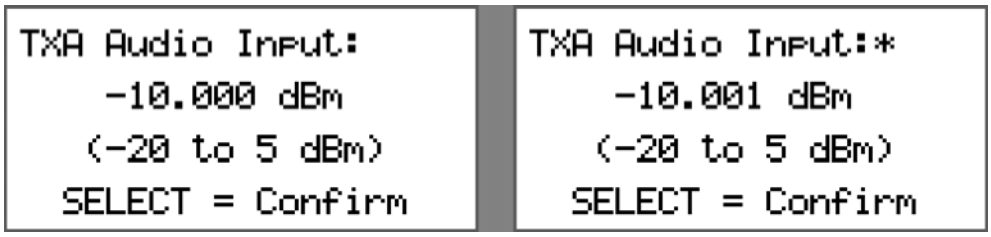
```
TXA 10MHz Cal:
     Skew: -720
  Press SELECT to
 start the routine
```

Refer to the "Calibrating 10 MHz reference" procedure in Section 4, "Non-standard maintenance" to perform the calibration.

TXA Audio input

When you select TXA Audio Input from the Calibration menu (see Figure 2.57) you can calibrate transmitter A's audio input level.

Figure 2.57: TXA Audio Input Screens

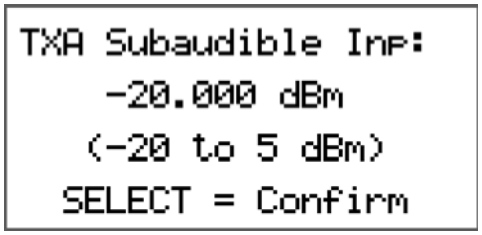


To calibrate transmitter A's audio input level, use the UP and DOWN buttons to increase or decrease the audio input level as desired, noting the range indicated. Note that an increase or decrease from the previously stored value will cause an asterisk to appear at the end of the screen's top line (see Figure 2.57). Press SELECT to confirm the setting. To cancel the setting and revert back to the previously stored value, press BACK, then press BACK again to discard the changes and return to the previous menu.

TXA Subaudible input

When you select TXA Subaudible Inp from the Calibration menu (see Figure 2.58) you can calibrate transmitter A's sub-audible input level.

Figure 2.58: TXA Sub-audible Input Screens

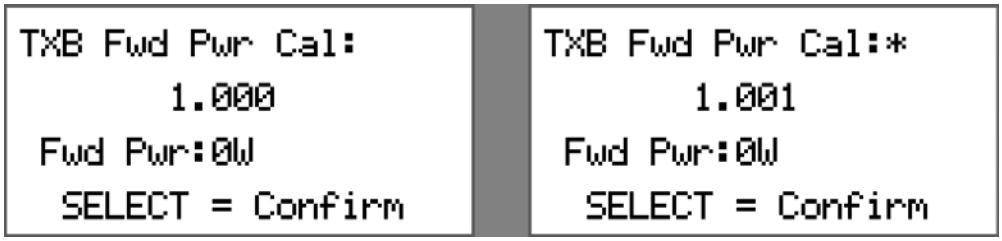


To calibrate transmitter A's sub-audible input level, use the UP and DOWN buttons to increase or decrease the sub-audible input level as desired, noting the range indicated. Press SELECT to confirm the setting. To cancel the setting and revert back to the previously stored value, press BACK, then press BACK again to discard the changes and return to the previous menu.

TXB Fwd Pwr Cal

When you select TXB Fwd Pwr Cal from the Calibration menu (see Figure 2.59) you can calibrate transmitter B's forward power level.

Figure 2.59: TXB Fwd Pwr Cal Screens

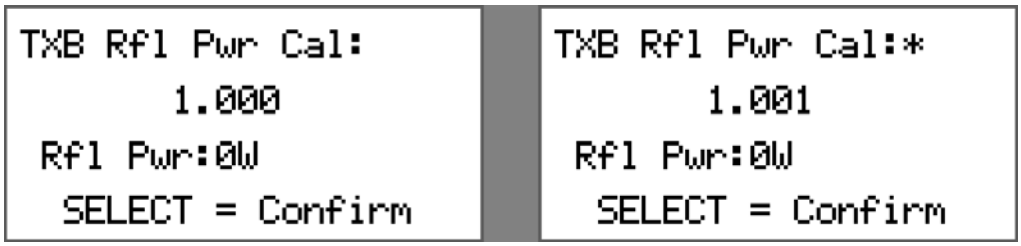


To calibrate transmitter B's forward power, use the UP and DOWN buttons to increase or decrease the scale factor as desired. Note that an increase or decrease from the previously stored value will cause an asterisk to appear at the end of the screen's top line (see Figure 2.59). Press SELECT to confirm the setting. To cancel the setting and revert back to the previously stored value, press BACK, then press BACK again to discard the changes and return to the previous menu. Note that the forward power (Fwd Pwr) displayed on the screen does not change until you press SELECT to confirm the setting. You can then re-adjust the scale factor, as necessary.

TXB Rfl Pwr Cal

When you select TXB Rfl Pwr Cal from the Calibration menu (see Figure 2.60) you can calibrate transmitter A's reflected power level.

Figure 2.60: TXB Rfl Pwr Cal Screens



To calibrate transmitter B's reflected power, use the UP and DOWN buttons to increase or decrease the scale factor as desired. Note that an increase or decrease from the previously stored value will cause an asterisk to appear at the end of the screen's top line (see Figure 2.60). Press SELECT to confirm the setting. To cancel the setting and revert back to the previously stored value, press BACK, then press BACK again to discard the changes and return to the previous menu. Note that the scale factor (Scale Fact) displayed on the screen does not change until you press SELECT to confirm the setting. You can then re-adjust the scale factor, as necessary.

TXB Preamp/IPA Bias

When you select TXB Preamp/IPA Bias from the Calibration menu (see Figure 2.61) you can calibrate transmitter B's pre-amplifier/IPA bias value.

Figure 2.61: TXB Preamp/IPA Bias Screen

```
TXB Preamp/IPA Bias

  Press SELECT to
 start the routine.
```

To calibrate transmitter B's pre-amplifier/IPA bias value, press the SELECT button to start a routine that will automatically callibrate the bias value. When the routine is complete, press BACK to return to the previous menu.

TXB 10MHz Cal

When you select TXB 10MHz Cal from the Calibration menu (see Figure 2.62) you can calibrate transmitter B's 10 MHz reference.

Figure 2.62: TXB 10MHz Cal Screens

```
TXB 10MHz Cal:
     Skew: -720
  Press SELECT to
 start the routine
```

Refer to the "Calibrating 10 MHz reference" procedure in Section 4, "Non-standard maintenance" to perform the calibration.

TXB Audio input

When you select TXB Audio Input from the Calibration menu (see Figure 2.63) you can calibrate transmitter B's audio input level.

Figure 2.63: TXB Audio Input Screens

```
TXB Audio Input:
    -10.000 dBm
   (-20 to 5 dBm)
  SELECT = Confirm
```

```
TXB Audio Input:*
    -10.001 dBm
   (-20 to 5 dBm)
  SELECT = Confirm
```

To calibrate transmitter B's audio input level, use the UP and DOWN buttons to increase or decrease the audio input level as desired, noting the range indicated. Note that an increase or decrease from the previously stored value will cause an asterisk to appear at the end of the screen's top line (see Figure 2.63). Press SELECT to confirm the setting. To cancel the setting and revert back to the previously stored value, press BACK, then press BACK again to discard the changes and return to the previous menu.

TXB Subaudible input

When you select TXB Subaudible Inp from the Calibration menu (see Figure 2.64) you can calibrate transmitter B's sub-audible input level.

Figure 2.64: TXB Sub-audible Input Screens

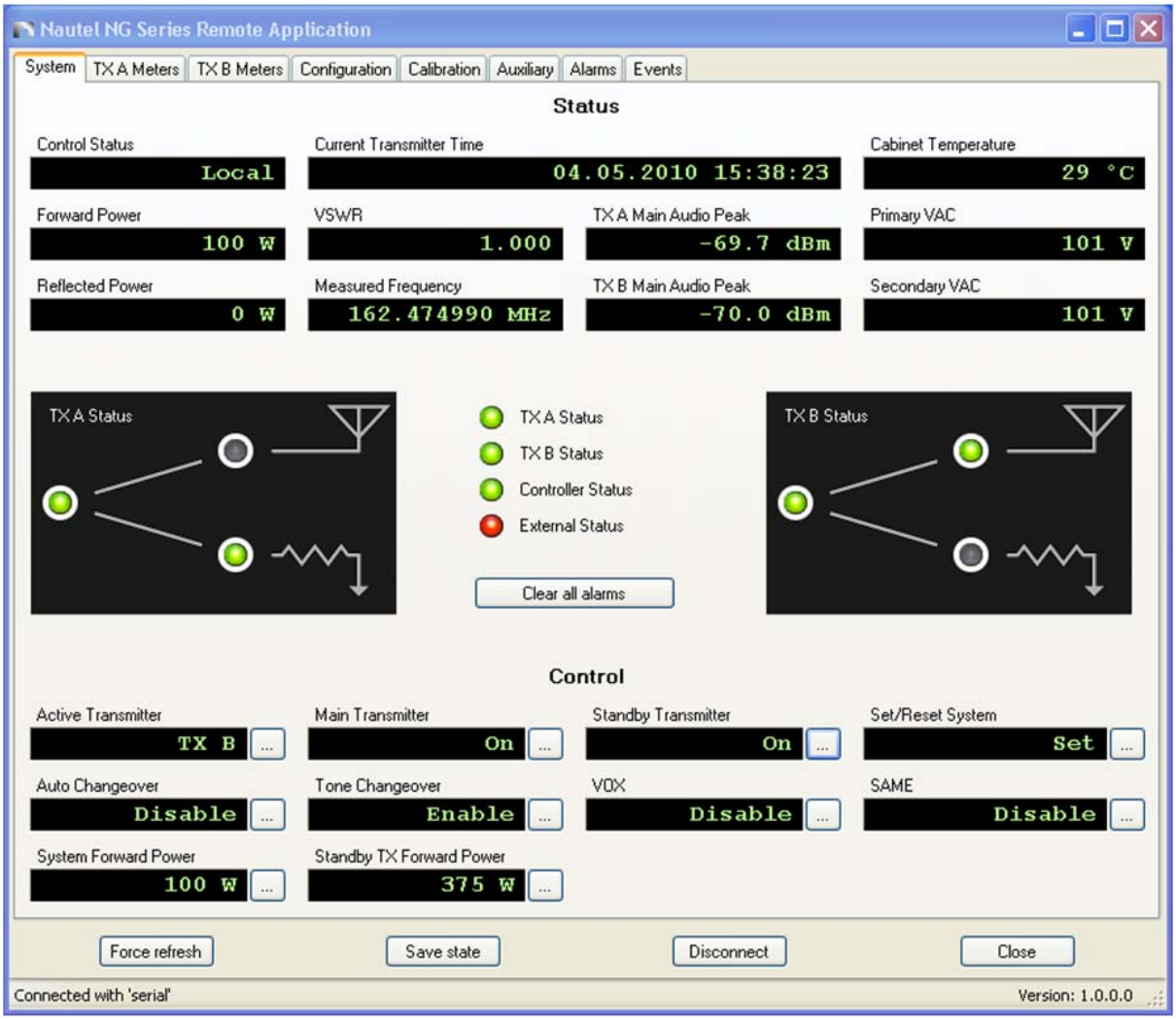
```
TXB Subaudible Inp:
    -20.000 dBm
   (-20 to 5 dBm)
  SELECT = Confirm
```

To calibrate transmitter B's sub-audible input level, use the UP and DOWN buttons to increase or decrease the sub-audible input level as desired, noting the range indicated. Press SELECT to confirm the setting. To cancel the setting and revert back to the previously stored value, press BACK, then press BACK again to discard the changes and return to the previous menu.

NG series remote application

You can completely control and monitor the NG1000 transmitter using a laptop or a PC that contains the "NG Remote Application" program (see Figure 2.65), available from Nautel.

Figure 2.65: NG Series Remote Application (System tab shown)



The majority of the features available through the system controller's front panel UI are also available using the remote application. Any changes made on the front panel UI or via the remote application will be visible on both displays.

The remote application has the following capabilities:

- Eight page tabs for system and transmitter (A and B) meters, configuration, calibration, auxiliary I/O, alarms and events
- Connects to the transmitter via modem (auto-negotiate: 300-19200 baud) or front panel RS-232 or USB connector
- Two levels of username/password access (user and technician) when connecting via modem; technician only access when connecting via RS-232 or USB
- Transmitter time and date can be entered manually or can automatically sync to PC
- Transmitter data, including meters, events, alarms and system state (including calibration), can be downloaded to CSV files
- Calibration files for the system controller and transmitters A and B can be saved to .CAL files
- Auxiliary I/O inputs and outputs can be identified and modified
- Remote digital output states can be viewed and modified

The remote application is structured using eight different tabs - System (shown in Figure 2.65 on page 2-47), TX A Meters, TX B Meters, Configuration, Calibration, Auxiliary, Alarms and Events - that are displayed along the top of any remote application page. Select a tab page by clicking on its tab name.

Each tab provides information about a specific transmitter function. See Table 2.2 for a brief description of each tab page and then to the the referenced page for a detailed description.

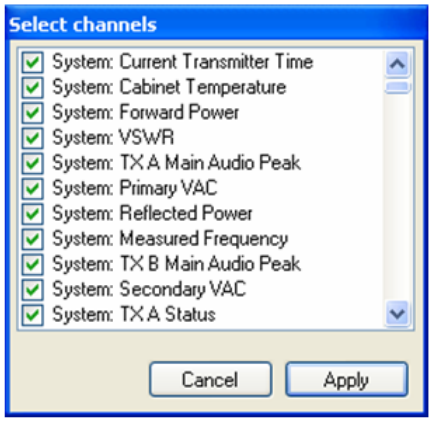
Most tab pages display fields that contain the state or metered value of various parameters. Fields that have an adjacent "change" button ... may be edited.

When the button is clicked, an appropriate pop-up window opens that allows the user to modify the field and then click Apply to confirm the change. Refer to the description of each tab page for specific examples of modifying fields.

There are four common control buttons that appear along the bottom of each page.

- **Force refresh**: re-transfers all displayed values from the transmitter site
- **Save state**: stores all alarms and levels available in the remote application to a CSV file. When clicked, the **Select channels** pop-up window (see Figure 2.66) appears. By default, all items are checked (selected). Uncheck any items you do not wish to save to the file. Click the **Apply** button to allow choosing a location to save the file.

Figure 2.66: Save state - Select channels window



- **Disconnect:** terminates the connection with the transmitter site and re-opens the Connection dialog box (see Figure 2.67 on page 2-50)
- **Close:** terminates the connection with the transmitter site and closes the remote application

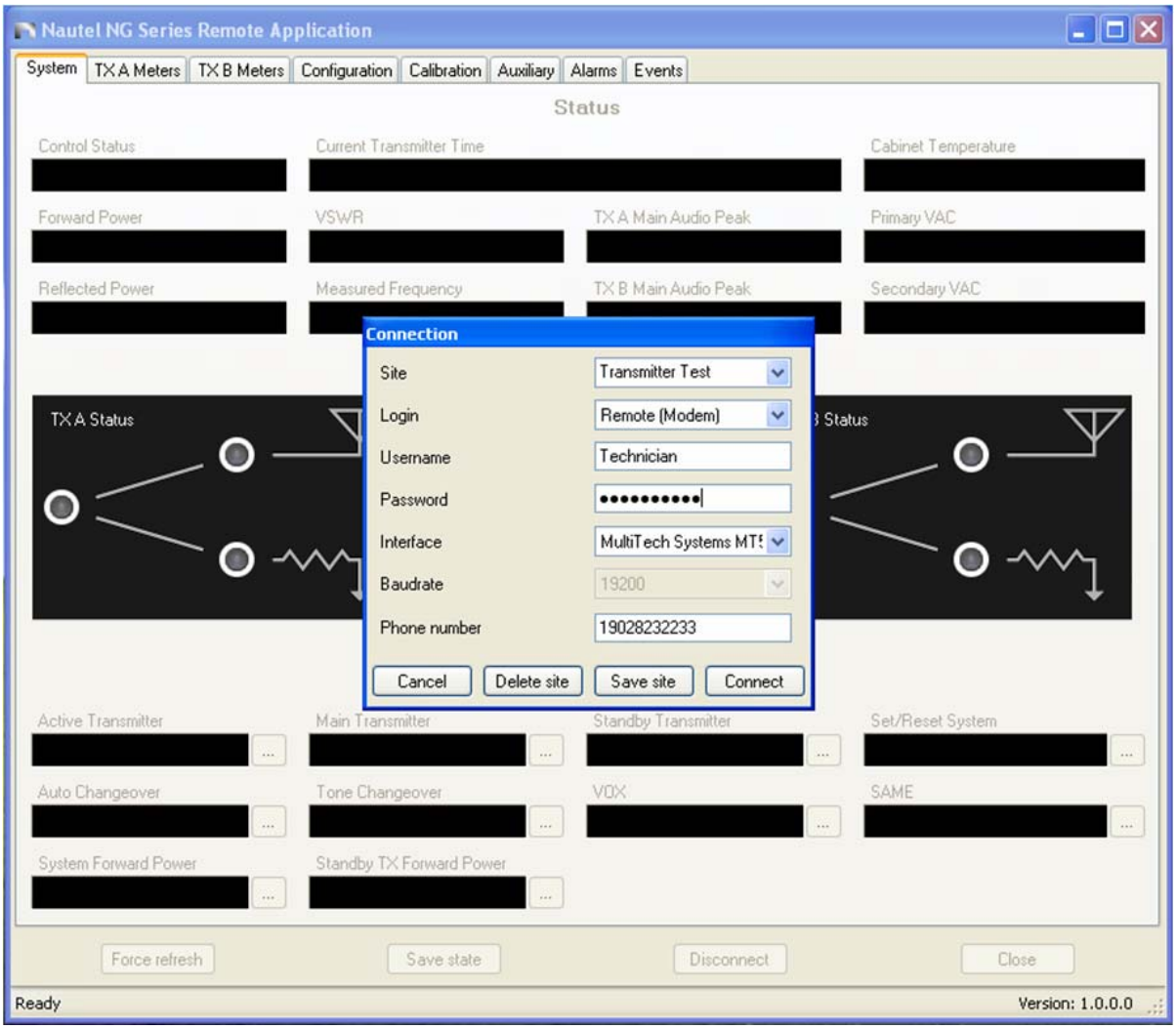
The text in the bottom, left corner indicates the connection status (e.g., *Connected with 'serial'*, *Connected* or *Ready*).

The text in the bottom, right corner indicates the version of the remote application.

Connecting to the remote application

The following information describes how to use the Connection dialog box to connect to the NG Series Remote Application.

Figure 2.67: Connection Dialog Box



To access the remote application using a PC or laptop, you must connect using the Connection dialog box (see Figure 2.67). This box appears during initial startup of the remote application or when the **Disconnect** button is clicked.

To launch the remote application for a pre-existing site, select the desired site from the drop-down menu and click the **Connect** button. To configure a site's connection details, enter the following information:

Site: When all other information is entered, you can click the Save site button and store the connection parameters with a Site name. After saving, the new site name will appear as an option in the Site drop-down menu. Use the Delete site button to delete the selected site from the previously saved location.

Login: Set the type of login to Remote (Modem) or Local (Serial), depending on the type of connection. A local login refers to a connection made to the RS-232 or USB connector on the front of the system controller.

Username: Enter the username for the site. When connecting via modem, there are two possible access usernames - User and Technician. When connecting via RS-232 or USB, there is only one possible access username - Technician.

Password: Enter the correct password (up to 12 alphanumeric characters) that corresponds to the user name. Passwords can only be set and changed by Technician level access personnel.

Interface: Select the appropriate interface from a drop-down list. When connecting via modem, a list of available system modems is displayed. When connecting via RS-232 or USB, a list of all available COM ports is displayed.

Baudrate: Select the baud rate connection speed of the associated interface device.

Phone number: When connecting via modem, enter the dial-in number to establish modem negotiations.

Connect: Click this button to begin connection to the selected site using the configured connection details.

NG series remote application tabs

Table 2.2 defines the tabs at the top of each NG Series Remote Application screen.

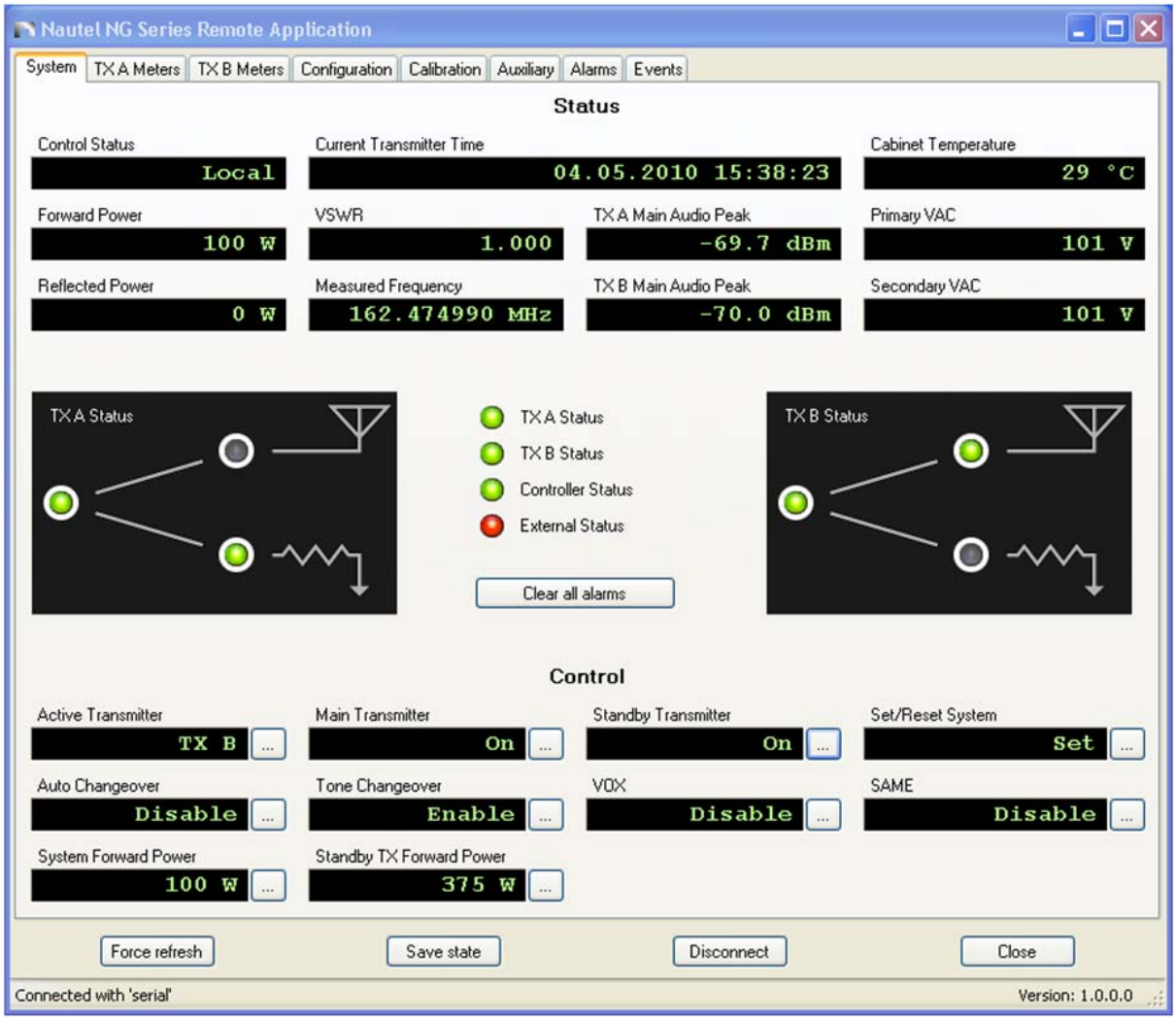
Table 2.2: NG Series Remote Application Tabs

AUI Page	Function	See Page
System	View system alarms and status and control various system parameters.	page 2-53
TX A Meters	View all critical parameter values for VHF transmitter A.	page 2-55
TX B Meters	View all critical parameter values for VHF transmitter B.	page 2-55
Configuration	Configure various transmitter site parameters, establish login information and upgrade firmware.	page 2-56
Calibration	Calibrate various parameters of the system controller and transmitters A and B.	page 2-59
Auxiliary	Configure all auxiliary I/O, including analog inputs, digital inputs and digital outputs.	page 2-61
Alarms	View all active alarms for the system, transmitters A and B and auxiliary I/O.	page 2-63
Events	View a list of all events in the system controller event log.	page 2-64

System tab

From the System tab (see Figure 2.68), you can view system alarms and status and control various system parameters.

Figure 2.68: System Tab



The System tab screen has two distinct sections - Status and Control.

The Status section contains fields that display all critical meter values for the system. Additionally, the status indicators from the system controller and VHF transmitter front panels are replicated and a Clear all alarms button is available.

The **Control** section contains fields that display the current state or value for critical settings and parameters. Users with appropriate access (see descriptions below) can edit fields in the **Control** section by clicking the ... button and noting the following options:

> *Note:*
> *Some fields can be modified by both User and Technician access. Other fields require Technician access. If a User attempts to edit a Technician-only access field, an error message will appear.*
>
> *The functionality of the control features of the remote application are identical to those of the front panel UI. Refer to Table 2.1 on page 2-6 to locate information on a particular control feature.*

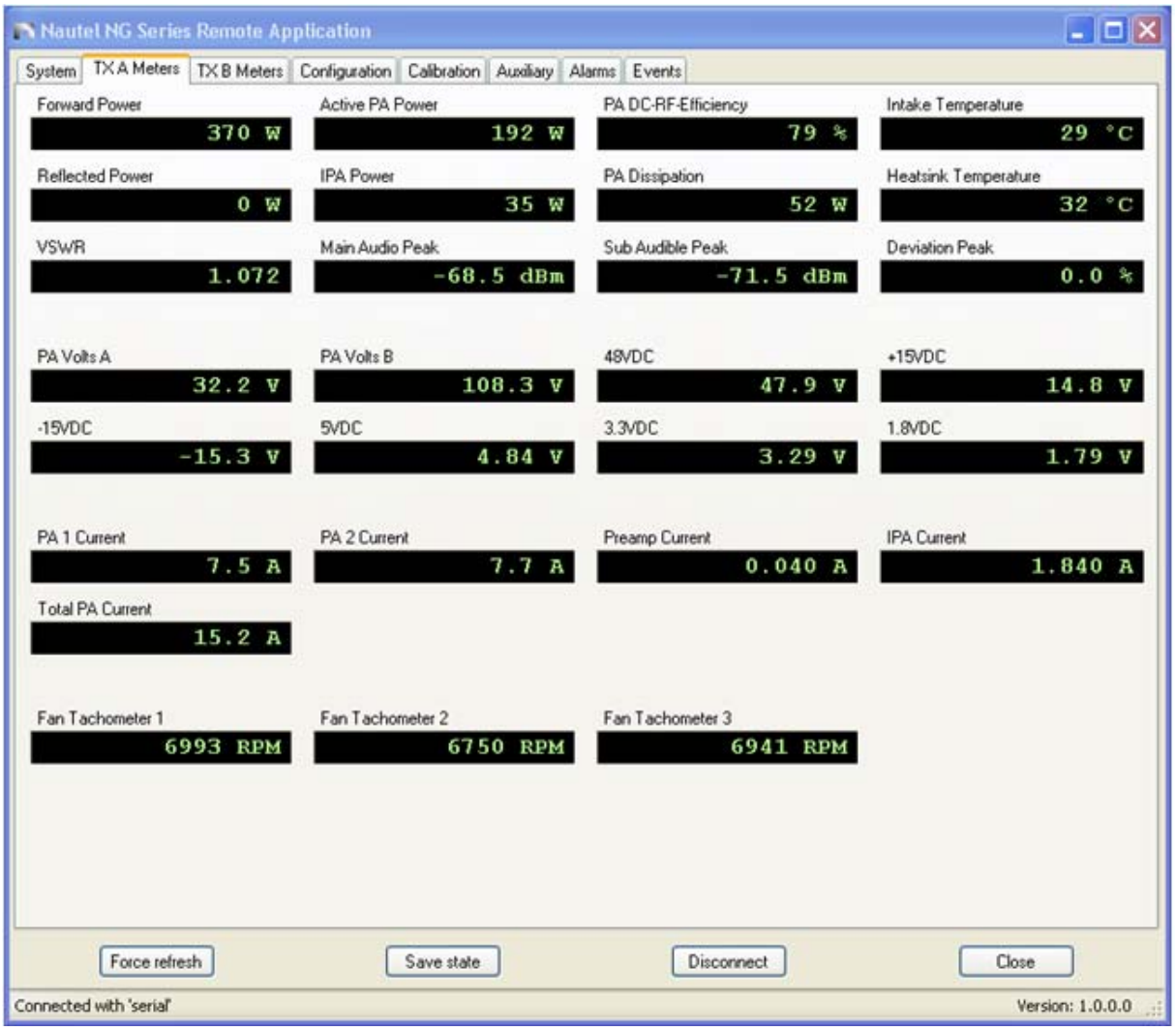
- **Active Transmitter**: select TX A or TX B (User or Technician)
- **Main Transmitter**: select On or Off (User or Technician)
- **Standby Transmitter**: select On or Off (User or Technician)
- **Set/Reset System**: select Set or Reset (Technician only)
- **Auto Changeover**: select Enable or Disable (User or Technician)
- **Tone Changeover**: select Enable or Disable (User or Technician)
- **VOX**: select Enable or Disable (Technician only)
- **SAME**: select Enable or Disable (User or Technician)
- **System Forward Power**: enter value between 200 W and 1000 W (User or Technician)
- **Standby Forward Power**: enter value between 300 W and 1250 W (Technician only)

TX A or B Meters tab

From the **TX A Meters** or **TX B Meters** tab (see Figure 2.69), you can view all critical parameter values for VHF transmitter A or B. All fields are for viewing purposes only.

> *Note:*
> *The measured values in the remote application are identical to those on the front panel UI.*

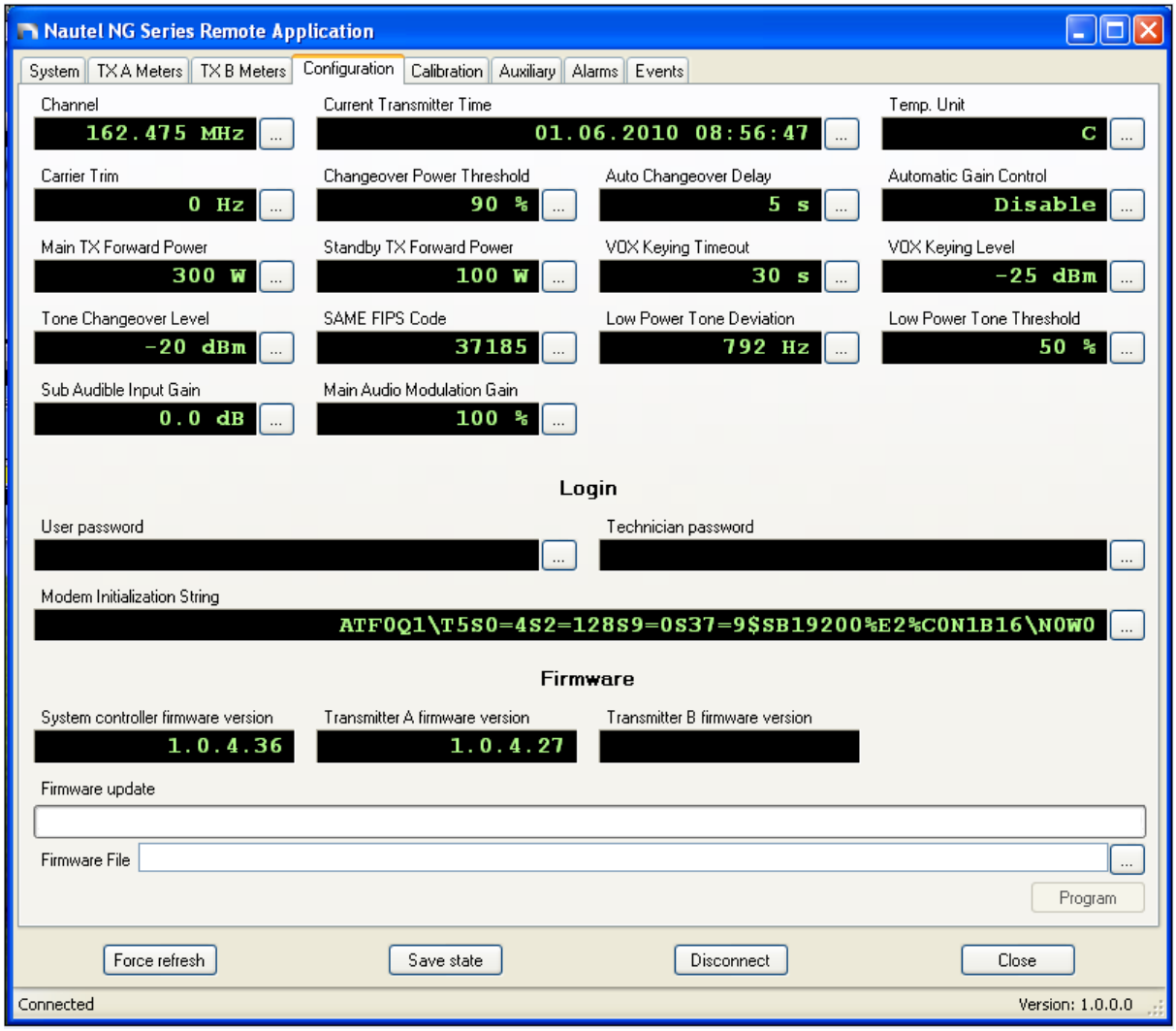
Figure 2.69: TX Meters Tab



Configuration tab

From the **Configuration** tab (see Figure 2.70), you can configure various transmitter site parameters, establish or change login information and upgrade firmware.

Figure 2.70: Configuration Tab



The Configuration tab screen has three distinct sections - configurable system parameters, **Login** and **Firmware**.

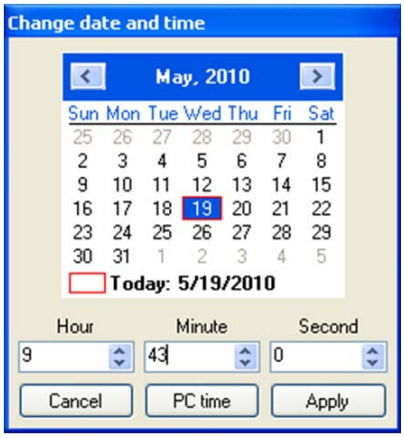
The upper section contains fields that display various system parameters and their current values. Users with appropriate access (see descriptions below) can edit fields in this section by clicking the ... button and noting the following options:

> *Note:*
> *Some fields can be modified by both User and Technician access. Other fields require Technician access. If a User attempts to edit a Technician-only access field, an error message will appear.*
>
> *The setting options for configurable parameters in the remote application are identical to those of the front panel UI. Refer to Table 2.1 on page 2-6 to locate information on a particular control feature.*

- **Channel** (User or Technician): select 162.400 MHz, 162.425 MHz, 162.450 MHz, 162.475 MHz, 162.500 MHz, 162.525 MHz or 162.550 MHz

- **Current Transmitter Time** (User or Technician): When clicked, the **Change date and time** window (see Figure 2.71) appears. Use the calendar to select the month and day and adjust the Hour, Minute and Second fields for the desired time. Click the **PC time** button if you wish to synchronize your transmitter system time to the connected PC. Click the **Apply** button to confirm the changes.

Figure 2.71: Change date and time window



- **Temp. Unit** (Technician only): select C (Celsius) or F (Fahrenheit)

- **Carrier Trim** (Technician only): enter a value between -1000 Hz and 1000 Hz

- **Changeover Power Threshold** (Technician only): enter a value between 0 % and 100%

- **Auto Changeover Delay** (Technician only): enter a value between 0 s and 20 s

- Automatic Gain Control (Technician only): select Enable or Disable
- Main TX Forward Power (User or Technician): enter a value between 200 W and 1000 W
- Standby TX Forward Power (Technician only): enter value between 300 W and 1250 W
- VOX Keying Timeout (Technician only): enter a value between 10 s and 60 s
- VOX Keying Level (Technician only): enter a value between -50 dBm and -25 dBm
- Tone Changeover Level (Technician only): enter a value between -50 dBm and -25 dBm
- SAME FIPS Code (User or Technician): enter a value between 0 and 999999
- Low Power Tone Deviation (Technician only): enter a value between 0 Hz and 5000 Hz
- Low Power Tone Threshold (Technician only): enter a value between 25% and 75%
- Sub Audible Input Gain (Technician only): enter a value between -6.0 dBm and 6.0 dBm
- Main Audio Modulation Gain (Technician only): enter a value between 10% and 100%

The Login section contains fields for login specific information. Users with Technician level access can edit fields in the Login section by clicking the ... button and noting the following options:

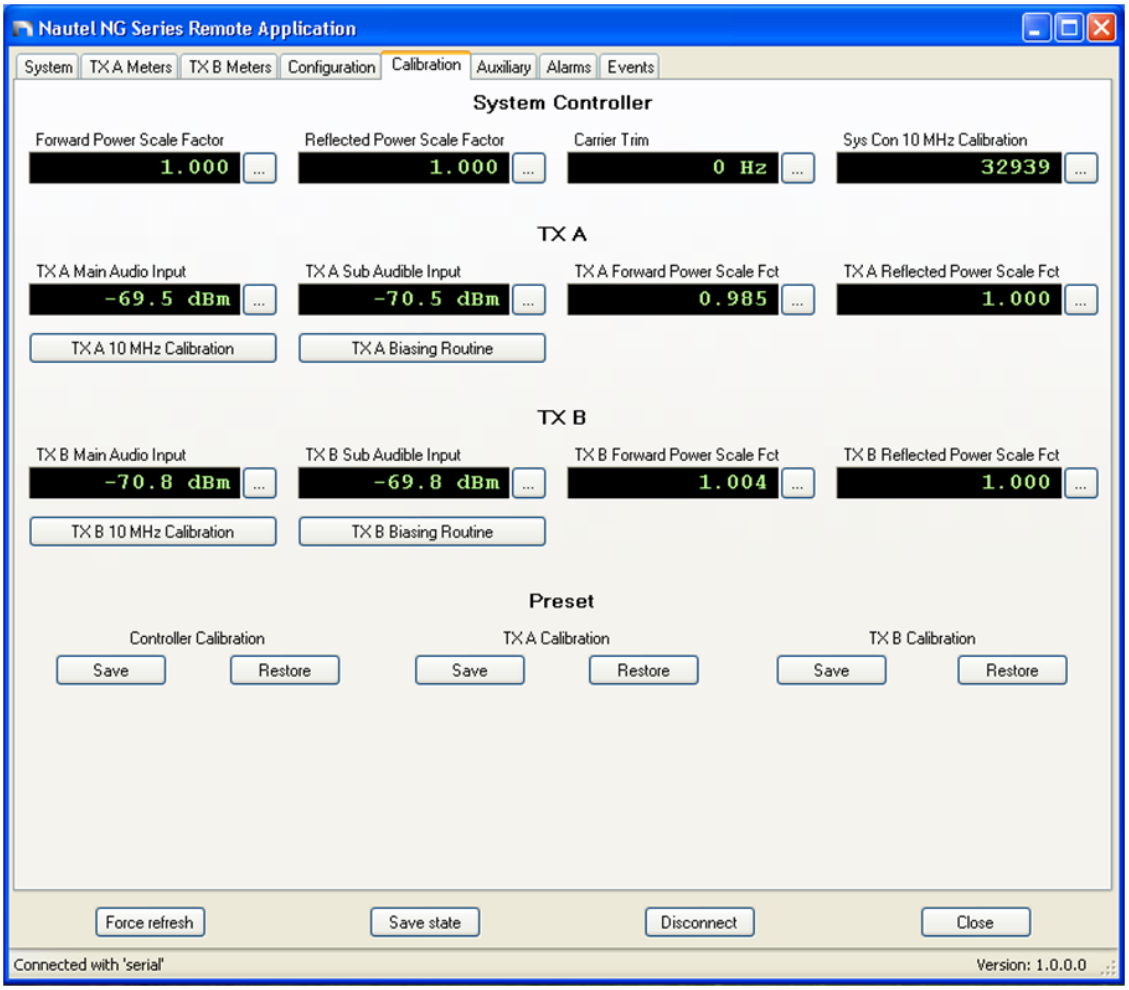
- User password: A prompt appears to "Insert new value for User password". Change as desired. The password is not displayed in any field within the remote application so it is recommended it be recorded and stored appropriately.
- Technician password: A prompt appears to "Insert new value for Technician password". Change as desired. The password is not displayed in any field within the remote application so it is recommended it be recorded and stored appropriately.
- Modem Initialization String: Enter the appropriate information.

The Firmware section contains fields that display the current firmware version for the system controller, transmitter A and transmitter B. The section also displays a Firmware update status bar and Update button that are used to perform software upgrades, for use by Technician level access only via the RS-232 or USB connection only. Upgrades via modem are prohibited. Refer to "Upgrading software" on page 4-1 of Section 4, "Non-standard maintenance" for detailed instructions to upgrade software.

Calibration tab

From the **Calibration** tab (see Figure 2.72), personnel with Technician level access can calibrate various parameters of the system controller and transmitters A and B. Calibration must be performed via RS-232 or USB connection. With the exception of calibrating the carrier trim value, calibration via modem is prohibited.

Figure 2.72: Calibration Tab



The Calibration tab screen has four distinct sections - **System Controller**, **TX A**, **TX B** and **Preset**.

> *Note:*
> *The setting options for calibration factors and values parameters in the remote application are identical to those of the front panel UI. Refer to Table 2.1 on page 2-6 to locate information on a particular calibration.*

The System Controller section contains fields that display various calibration factors or values for system parameters. Users with Technician level access can edit fields in this section by clicking the ... button and noting the following options:

- Forward Power Scale Factor: enter a scale factor between 0.500 and 2.000
- Reflected Power Scale Factor: enter a scale factor between 0.500 and 2.000
- Carrier Trim: enter a value between -1000 Hz and 1000 Hz
- Sys Con 10 MHz Calibration: enter a value between 0 and 65535. Refer to "System controller 10 MHz calibration" on page 4-9 of Section 4, "Non-standard maintenance" for more information.

The TX A and TX B sections each contains fields that display various calibration factors or values and two calibration routine buttons for transmitter A or B parameters. Users with Technician level access can edit fields in this section by clicking the ... button and noting the following options:

- TX A or B Main Audio Input: enter a value between -20.0 and 5.0 dBm
- TX A or B Sub Audible Input: enter a value between -20.0 and 5.0 dBm
- TX A or B Forward Power Scale Fct: enter a scale factor between 0.500 and 2.000
- TX A or B Reflected Power Scale Fct: enter a scale factor between 0.500 and 2.000
- TX A or B 10 MHz Calibration button: when clicked, starts a calibration routine on transmitter A or B's 10 MHz reference. Refer to "Transmitter A and B 10 MHz calibration" on page 4-9 of Section 4, "Non-standard maintenance" for more information.
- TX A or B Biasing Routine button: when clicked, starts an automatic biasing routine on transmitter A or B's pre-amp/IPA PWB.

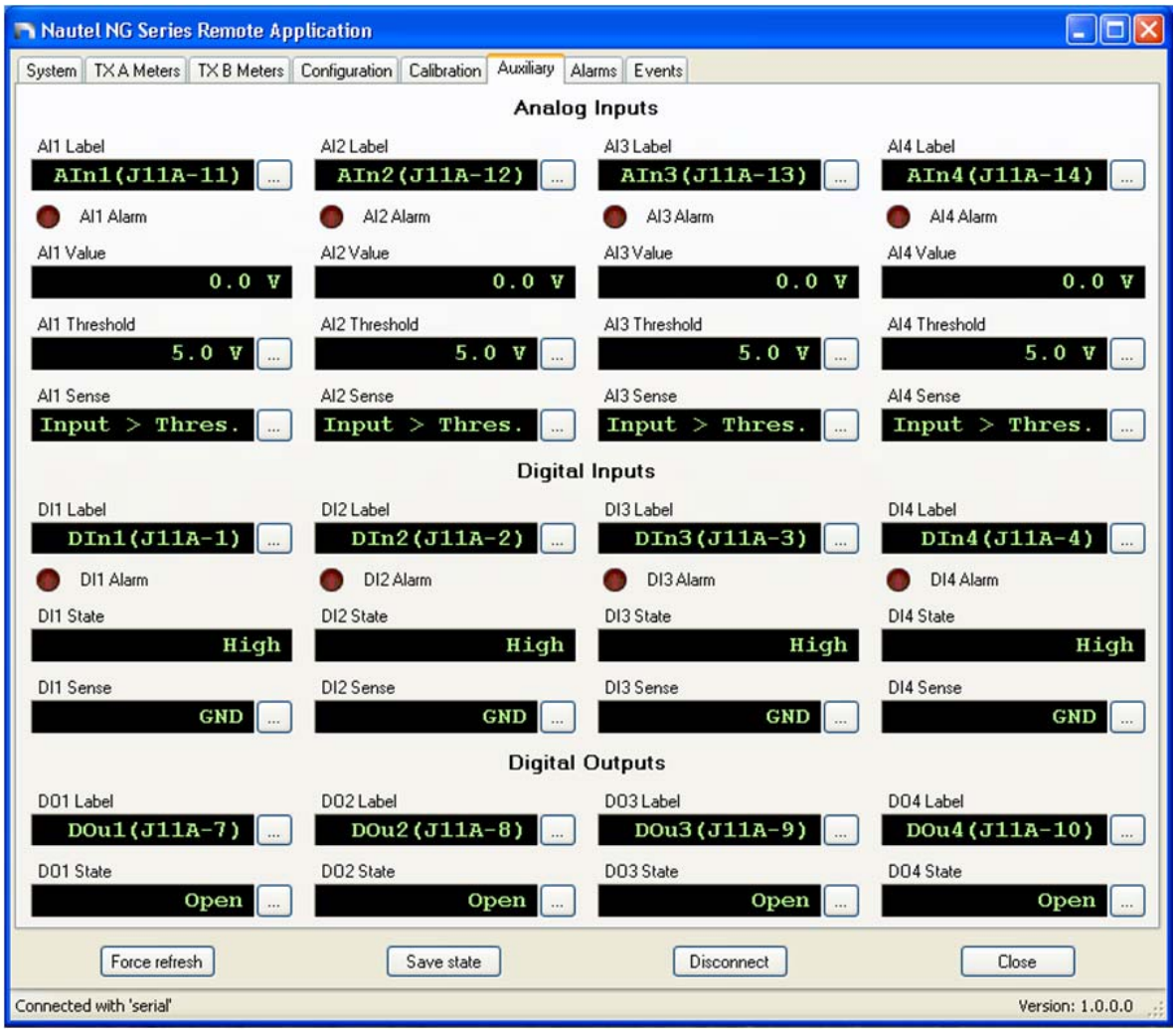
The Preset section contains buttons that allow the user to save and restore calibration data for all three system components (Controller, TX A and TX B).

- Save button: When clicked, a pop-up window appears that prompts the user to choose a filename and location for the saved file (.cal file type).
- Restore button: When clicked, a pop-up window appears that prompts the user to choose a filename and location for the restore file. A second pop-up window appears to allow selection of the parameters to be restored.

Auxiliary tab

From the **Auxiliary** tab (see Figure 2.73), any user can configure the analog inputs and digital inputs/outputs for the transmitter system.

Figure 2.73: Auxiliary Tab



The Auxiliary tab screen has three distinct sections - **Analog Inputs**, **Digital Inputs** and **Digital Outputs**.

> *Note:*
> *With the exception of the ability to label input and outputs, the setting options for auxiliary inputs and outputs in the remote application are identical to that described for the front panel UI. Refer to Auxiliary I/O on page 2-34 for more information.*

The **Analog Inputs** section contains fields that display labels, current values, thresholds and input sense settings for the four analog inputs. There is also an alarm for each input that indicates if the analog input has tripped its configured sense threshold. Personnel with User or Technician level access can edit fields in this section by clicking the ... button and noting the following options:

- **AI1 through 4 Label**: enter a label (maximum 13 characters) for the analog input
- **AI1 through 4 Alarm**: indicates if the analog input level is outside the pre-defined threshold level and input sense setting
- **AI1 through 4 Value**: displays the current level of the analog input
- **AI1 through 4 Threshold**: enter a value between 0.0 V and 12.0 V
- **AI1 through 4 Sense**: select either **Input > Thres.** or **Input < Thres.**

The **Digital Inputs** section contains fields that display labels, state and input sense settings for the four digital inputs. There is also an alarm for each input that indicates if the digital input switches to its alarm sense logic. Personnel with User or Technician level access can edit fields in this section by clicking the ... button and noting the following options:

- **DI1 through 4 Label**: enter a label (maximum 13 characters) for the digital input
- **DI1 through 4 Alarm**: indicates if the digital input logic level is at the pre-defined input sense setting
- **DI1 through 4 State**: displays the current state of the digital input
- **DI1 through 4 Sense**: select either **Gnd** or **Open**

The **Digital Outputs** section contains fields that display labels and state settings for the four digital outputs. Personnel with User or Technician level access can edit fields in this section by clicking the ... button and noting the following options:

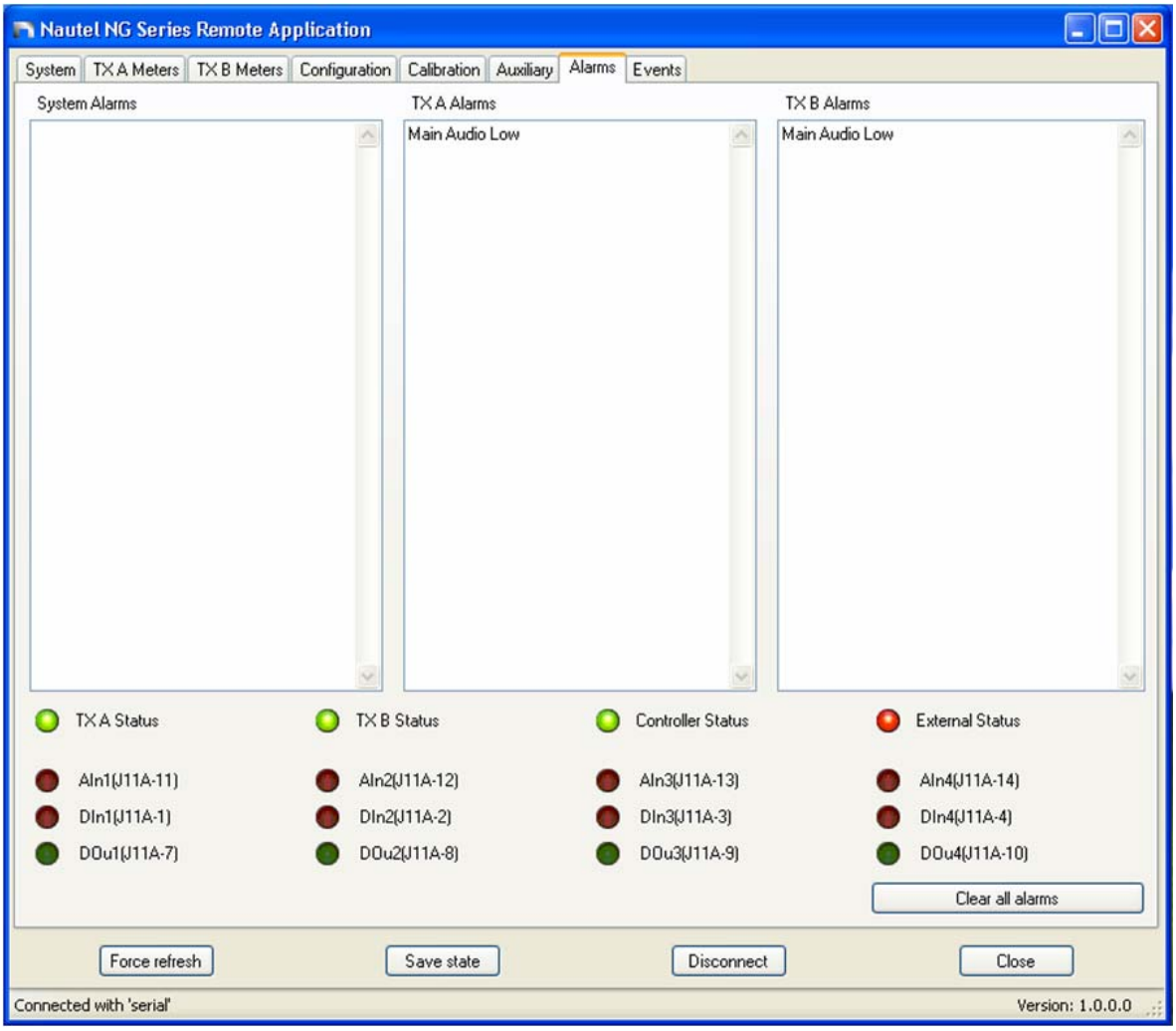
- **DO1 through 4 Label**: enter a label (maximum 13 characters) for the digital output
- **DO1 through 4 State**: select either **Gnd** or **Open**. This will be the state applied to the digital output.

Alarms tab

From the **Alarms** tab (see Figure 2.74), you can view all active alarms for the system, transmitters A and B and auxiliary input and outputs. Information on this screen is for viewing purposes only.

> *Note:*
> *The system and transmitter (TX) alarms displayed in the remote application are identical to those on the front panel UI. The Auxiliary input/output alarms are identical to those on the Auxiliary tab's screen.*

Figure 2.74: Alarms Tab



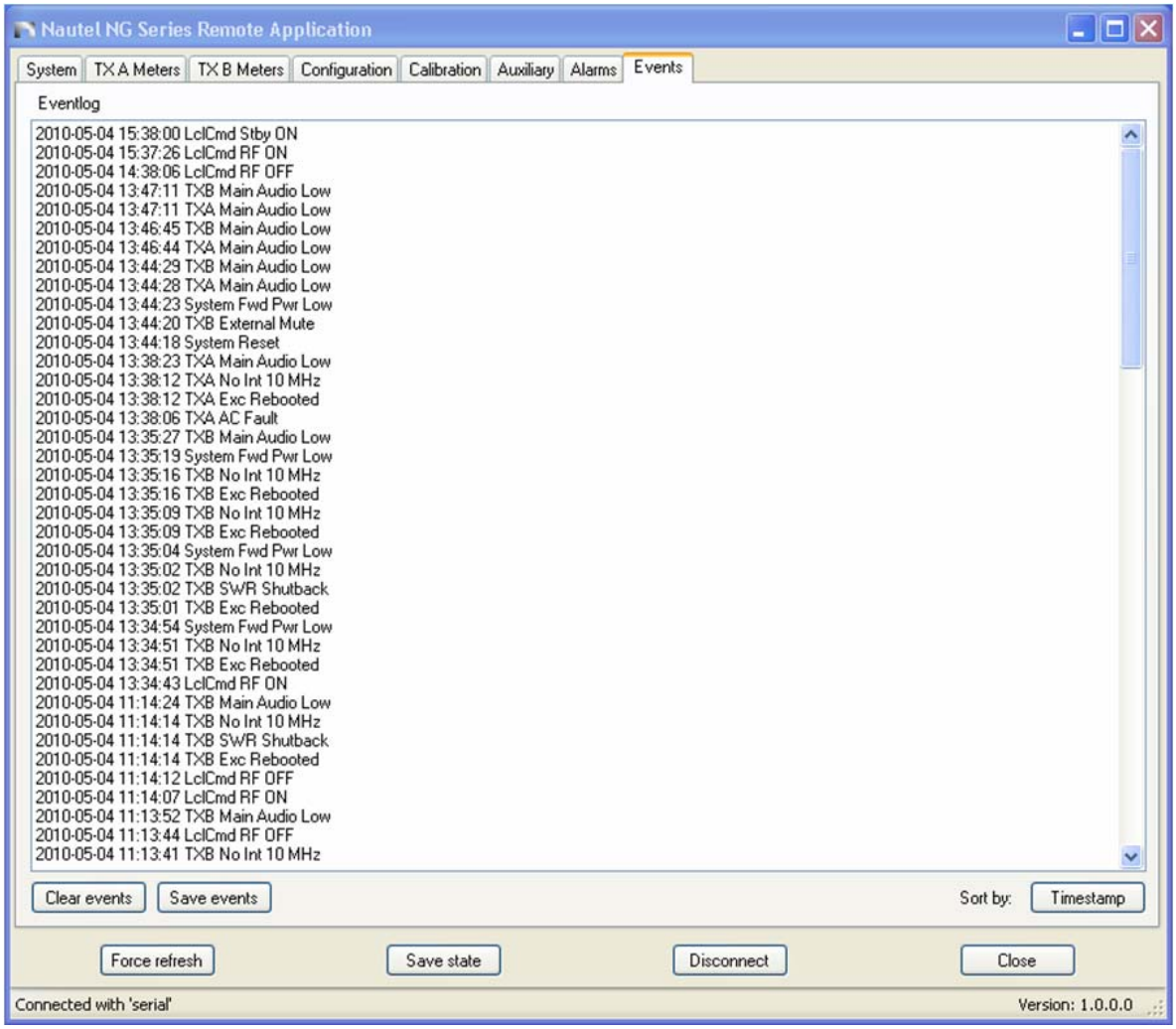
Personnel with Technician level access can use the **Clear all alarms** button to attempt to clear all alarms displayed. Any alarms not still active will be cleared.

Events tab

From the **Events** tab (see Figure 2.75), you can view a list of all events in the system controller event log. Information on this screen is for viewing purposes only.

> *Note:*
> *The events displayed in the remote application are identical to those on the front panel UI.*

Figure 2.75: Events Tab



If there are more events than can be displayed on one page, a scroll bar will appear on the right hand side of the screen.

Personnel with Technician level access can use the Clear events button to clear all events from the list. This also deletes all logs from the system controller's front panel UI.

Use the Save events button to save a copy of the events log as a CSV file. When you click the button, a pop-up window appears that allows the user to choose a file location for the saved events log.

Use the Sort by: button to sort the events by Timestamp, which sorts the events in order of time (most recent to oldest) or by Cause which sorts the events by type. If there are multiple events for a particular cause, that group of events will be sorted from most recent to oldest. The button will display the current sorting method.

Section 3: Routine maintenance

This section provides instructions for performing routine maintenance on the NG1000 transmitter. This section includes the following topics:

- Scheduled maintenance
  - Cleaning
  - Checking hardware - see page 3-2
- Replacing a damaged air filter - see page 3-2
- Performing on-air checks - see page 3-4
- Inspecting lightning protection systems - see page 3-5
- Maintenance Data Log - see page 3-6

Scheduled maintenance

Scheduled maintenance consists of performing a visual inspection of the NG1000 transmitter system at scheduled intervals. The recommended minimum time between scheduled maintenance visits is three months. Local operating and environmental conditions may dictate more frequent visits, while in remote sites less frequent visits may be acceptable. Experience and system reliability will determine the most practical schedule for a specific installation. The user should adopt a maintenance data log similar to the example shown in Table 3.1 on page 3-6.

Cleaning

At a minimum, follow the cleaning procedures listed below.

> *Note:*
> *A site located in a dirty area using open-air cooling requires more extensive cleaning than a site located in a clean area using closed-air cooling.*

1. Clean the air filter in the front of each VHF transmitter. Remove each air filter (see Figure 3.1 on page 3-3) by removing the front panel grill section that is secured with four M4 hex screws. Use the HY88 hex tool provided in the ancillary kit to loosen the screws. Remove each filter media, rinse it with water, then re-install the air filter. If an air filter becomes damaged, replace it as soon as possible. See "Replacing a damaged air filter" on page 3-2.

2. Clean the NG1000 cabinet using a vacuum cleaner and a soft-bristle brush to remove loose dirt. Clean, damp rags should be used to remove dirt that cannot be removed with a vacuum cleaner. Never use compressed air to clean the NG1000 cabinet.

Checking hardware

All hardware must be checked at least once a year. Thermal cycling from turning the NG1000 on and off will require more frequent checks.

- To prevent damaging the hardware, ensure that the proper sized tools are used. In most cases, hardware is metric.
- A mounting surface or terminal that has changed colour indicates a loose hardware connection.

Replacing a damaged air filter

If an air filter becomes damaged, replace it as follows:

Parts and tools

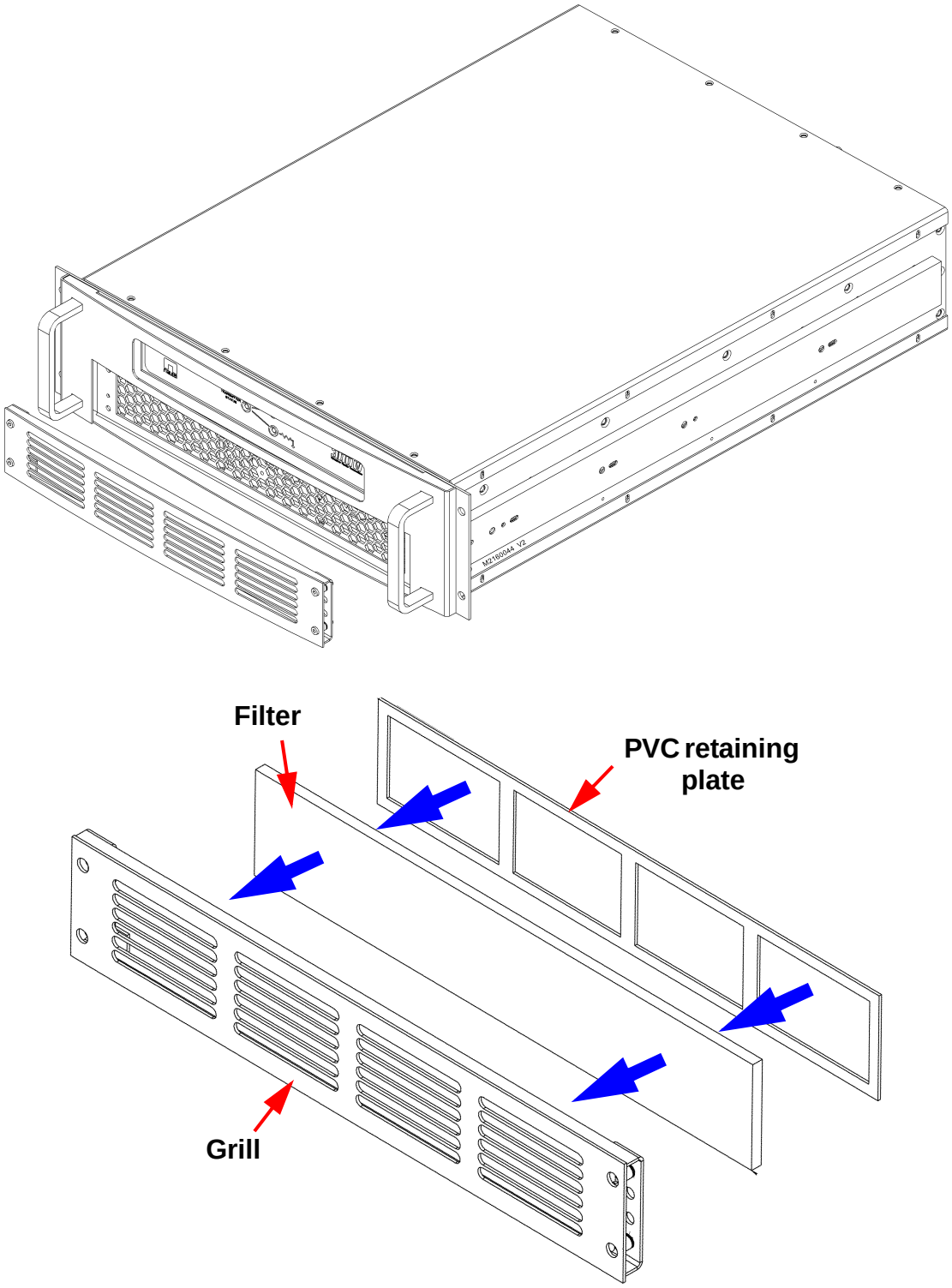
- Replacement air filter
- T-handle hex tool, 3 mm (Nautel Part # HY88, provided in ancillary kit)

Procedure

Take the following steps to replace an air filter in one of the VHF transmitters (see Figure 3.1 on page 3-3):

1. Order a new filter, as necessary. You can order a standard sized filter from Nautel (Nautel Part # HAM75) or from the manufacturer (Universal Air Filter, 25 PPI Quadrafoam Media, Black) or obtain one from any local supplier of HVAC supplies.
2. Gain access to the old filter by removing the front panel grill section secured with four captive M4 hex screws. Use the HY88 hex tool from the ancillary kit to loosen the screws.
3. Temporarily remove the PVC retaining plate from the grill section and then remove the old filter.
4. Insert the new air filter, re-insert the PVC retaining plate into the grill section and then re-install the front panel grill.
5. Dispose of the damaged air filter.

Figure 3.1: Air Filter Cleaning or Replacement



Performing on-air checks

You should check the on-air quality of each system transmitter's signal every 12 months, or as required by law. This check can be done locally at the site, or remotely from another location in the system's coverage area.

Parts and tools

- High-quality receiver compatible with the modulation scheme used on the transmitters.

Procedure

1. Monitor each VHF transmitter's signal to identify any problems:
   - Use a high quality receiver to monitor audio quality.
   - Check the spectral output of the VHF transmitter by connecting a spectrum analyzer to the directional coupler's RF MONITOR BNC output (A4A1J3).
   - Take a portable receiver to different locations in your coverage areas to identify any problems with the antenna pattern or areas of localized interference.
2. Obtain the *Critical Parameters* sheet that was completed during factory testing and provided with the transmitter. Use the system controller's front panel UI to compare current transmitter settings with those on the *Critical Parameters* sheet.

> *Note:*
> *Nautel completes the Critical Parameters sheet with each transmitter terminated into a precision 50 ohm load. Measurements made on site into a dummy load or station antenna may not yield the same readings. Slight variances are acceptable. The data is provided as a troubleshooting aid.*

Inspecting lightning protection systems

Every six months you should inspect the lightning protection systems that are connected to your NG1000 transmitter system, ac power, RF feedline, antenna and tower periodically to ensure that your site, equipment and personnel are fully protected.

Parts and tools

Depends on your site, but may include a digital multimeter, ladder, binoculars, and continuity tester.

Procedure

Perform the following steps to test the lightning protection systems:

1. Test the continuity between ground available at your lightning arrestor and ground available on various pieces of equipment connected to the transmitter system. Ensure that there is no resistance (near 0 ohms, accounting for multimeter test lead resistance).

2. Physically inspect the lightning arrestor to ensure that all connections are tight. Also look for any sign of scoring or burning (indications of possible damage from earlier lightning strikes).

3. Using binoculars, visually inspect the hardware on the tower. Look for charred, broken or frayed connections and corrosion.

Table 3.1: Maintenance Data Log

Test / Action	Date:		Date:		Date:	
	Primary	Secondary	Primary	Secondary	Primary	Secondary
Corrective Action (Y or N)						
Clean Air Filter (Y or N)						
Visual Inspection (Y or N)						
Check Hardware (Y or N)						
Check LEDs (Y or N)						
Forward Power (W)						
Ref;ected Power (W)						
VSWR						
Total PA Current (A)						
PA Voltage (V)						
Power Amp # 1 Current (A)						
Power Amp # 2 Current (A)						
Power Amp # 3 Current (A)						
Power Amp # 4 Current (A)						
Power Amp # 5 Current (A)						
Power Amp # 6 Current (A)						
Power Amp # 7 Current (A)						
Power Amp # 8 Current (A)						
Intake Temperature (deg C or F)						
Phone Line Audio (V RMS)						
Audio Quality Check (Y or N)						

Table 3.1: Maintenance Data Log

Test / Action	Date:		Date:		Date:	
	Primary	Secondary	Primary	Secondary	Primary	Secondary
Modulation Deviation (+ or - kHz)						
Frequency Error (+ or - Hz)						
Reset Frequency? (Y or N)						
Frequency Reset To (+ or - Hz)						
Direct Computer Connect (Y or N)						
Remote Computer Connect (Y or N)						
AC Mains (V AC)						

Section 4: Non-standard maintenance

This section provides instructions for performing non-standard maintenance on the NG1000 transmitter. This section includes the following procedures:

- Upgrading software
- Changing frequency - see page 4-5
- Calibrating 10 MHz reference - see page 4-9

Upgrading software

Periodically, enhancements or improvements to the operating system are identified and made available. This procedure upgrades the system software using the "NG Series Remote Application". Perform a software upgrade, as required, as follows:

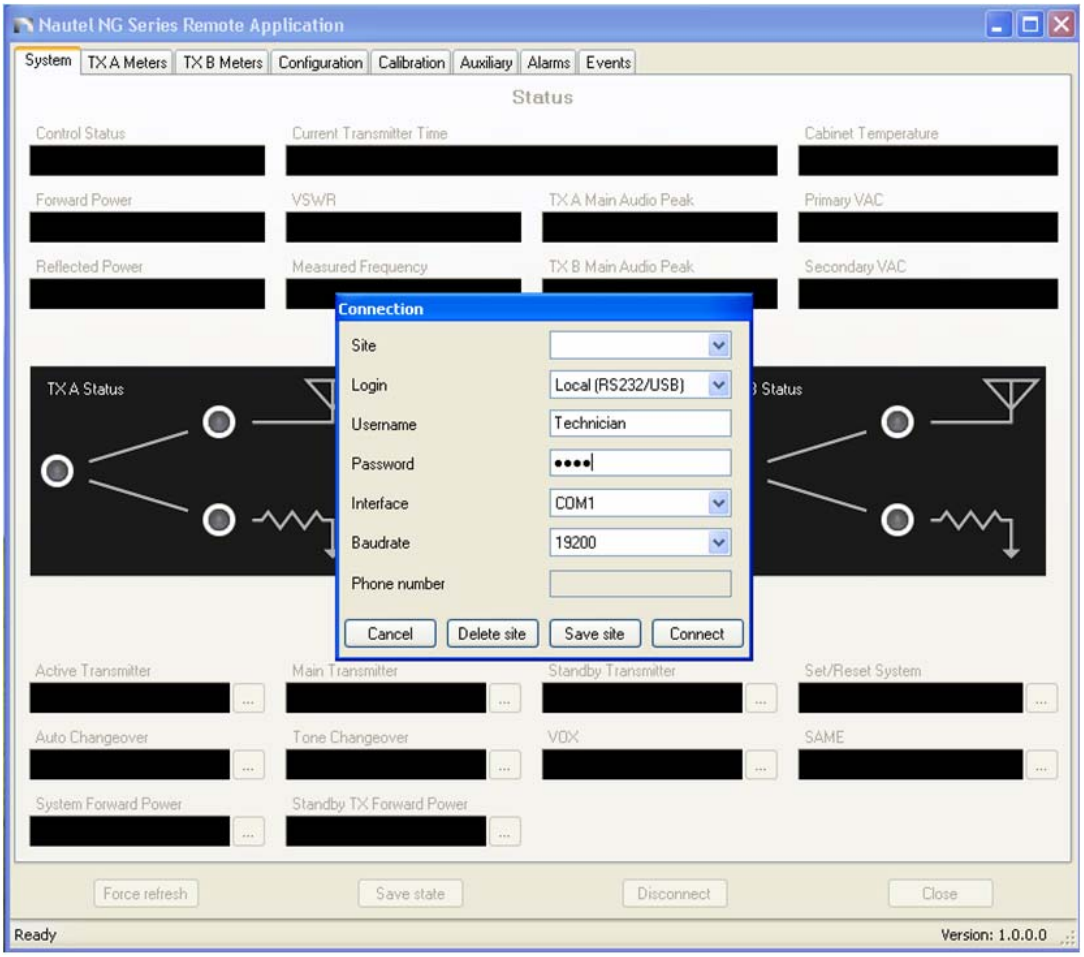
Parts and tools

- Laptop (i.e., local computer for direct connection) with "NG Series Remote Application" installed.
- Appropriate cable to connect between the laptop and the system controller's front panel USB or RS-232 (serial) port.
- Upgrade software (*.tgz file), provided via email, web download or flash card

Procedure

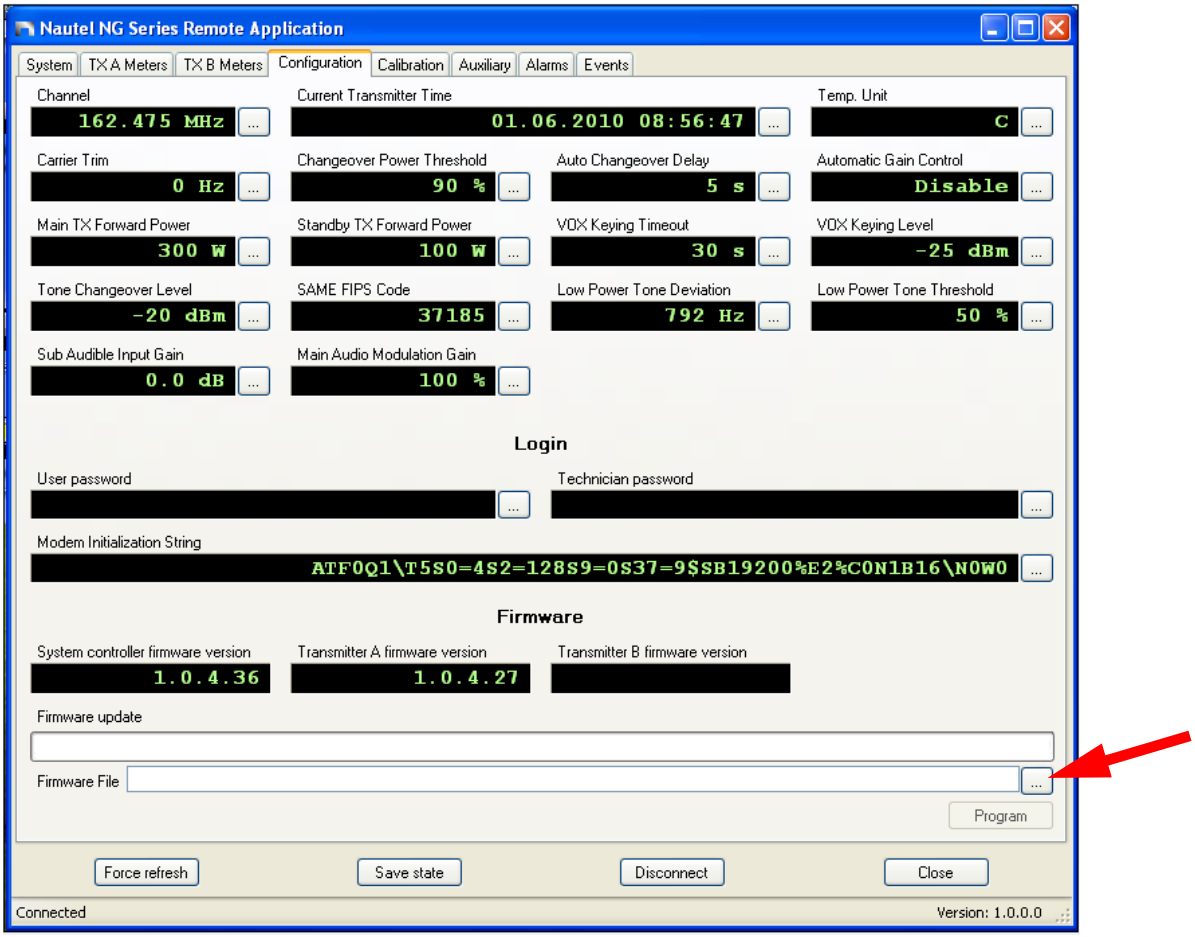
1. Determine the current software revision installed. Via the front panel screen, go to Main Menu -> About. Using the "NG Series Remote Application", go to the Configuration tab. The current firmware versions for the system controller and transmitters A and B will be displayed under the Firmware section.
2. Contact Nautel for the most recent software revision and receive a new compact flash card (which contains upgrade *.tgz file) or download *.tgz file to a laptop or network location.
3. Connect a laptop with "NG Series Remote Application" to the system controller's USB or RS-232 (serial) port.

Figure 4.1: NG Series Remote Application connection



4. On the laptop, run the "NG Series Remote Application" and click **Connect** (see Figure 4.1).

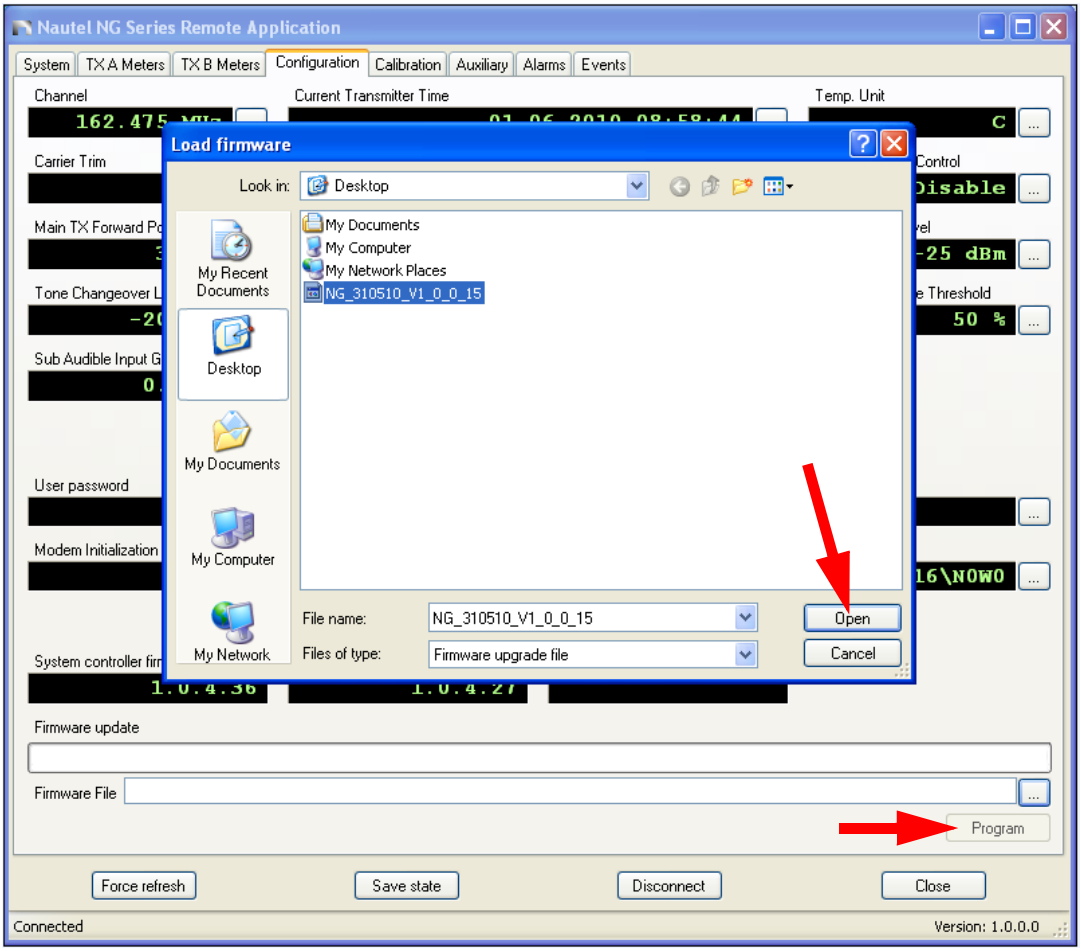
Figure 4.2: Configuration tab



5. Click on the **Configuration** tab (see Figure 4.2).

6. Click the ... button.

Figure 4.3: Browse and Load Firmware



7. Browse for the correct *.tgz* file (see Figure 4.3).

8. Select the correct *.tgz* file and click **Open**. The firmware filename will be displayed at the bottom of the screen. Click the **Program** button. The upgrade procedure will begin. The *Firmware update* status bar will indicate the progress of the upgrade.

9. Verify the updated firmware versions are displayed in the *Firmware* section. You can also view firmware versions on the system controller's UI using the **Main Menu** -> **About** screen.

10. When the upgrade has successfully completed, disconnect the system controller from the "NG Series Remote Application".

Changing frequency

Unless otherwise specified at the time of purchase, the NG1000's bandpass filter is factory tuned for a channel frequency of 162.475 MHz. If it is necessary to operate at a different channel frequency, perform the following procedure.

> *NOTE:*
>
> *Nautel recommends you use a calibrated network analyzer to tune the bandpass filter (see "Tuning with a network analyzer").*
>
> *If a network analyzer is not available, you can use a calibrated impedance meter that is capable of measuring VSWR or return loss, and a calibrated 50 ohm termination (see "Tuning with an impedance meter" on page 4-8).*
>
> *Refer to the appropriate tuning procedure for the test equipment being used.*

Test equipment and tools

- Network analyzer (recommended) or impedance meter
- Interconnect test cables and any required adapters
- 50-ohm termination, VSWR < 1.05:1, suitable for connection to the RF output (only required when tuning with impedance meter)
- 11/16" wrench

Tuning with a network analyzer

1. Turn off the transmitter system and disconnect (or disable) its ac power source(s).
2. Disconnect the P2 end of coaxial cable W7 from the input of the bandpass filter (U2-INPUT).
3. Disconnect the antenna feed cable from the transmitter system's RF output (A4J2).

4. Set up the network analyzer as follows:

   – Measurement Type: Thru Response (S21)
   – Center Frequency: site frequency
   – Span: 3 MHz
   – Scale: 2 dB/division
   – Reference Level: +2 dB
   – IF Bandwidth: 300 Hz

5. Connect the interconnecting test cables to each other and between the measurement ports of the network analyzer. Perform a thru response calibration.

   – Connect the first measurement cable between port 1 on the network analyzer and the bandpass filter's input (U2-INPUT).
   – Connect the second measurement cable between port 2 on the network analyzer and the RF output (directional coupler output A4J2).
   – Set marker 1 to the center frequency.
   – Set marker 2 to 1 MHz below the center frequency.
   – Set marker 3 to 1 MHz above the center frequency.

6. Loosen the brass locking nut at the bottom of the center tuning post using an 11/16" wrench.

7. Adjust the tuning knob (see Figure 4.4 on page 4-7) for minimum insertion loss (S21). You can increase the sweep rate by increasing the IF BW to 1000 Hz or more.

8. Re-tighten the brass locking nut. Ensure the filter remains tuned for minimum insertion loss. Loosen and re-adjust if necessary.

> *NOTE:*
> *When the filter is properly tuned, insertion loss (marker 1) should be less than -0.5 dB and markers 2 and 3 should be greater than 10 dB.*

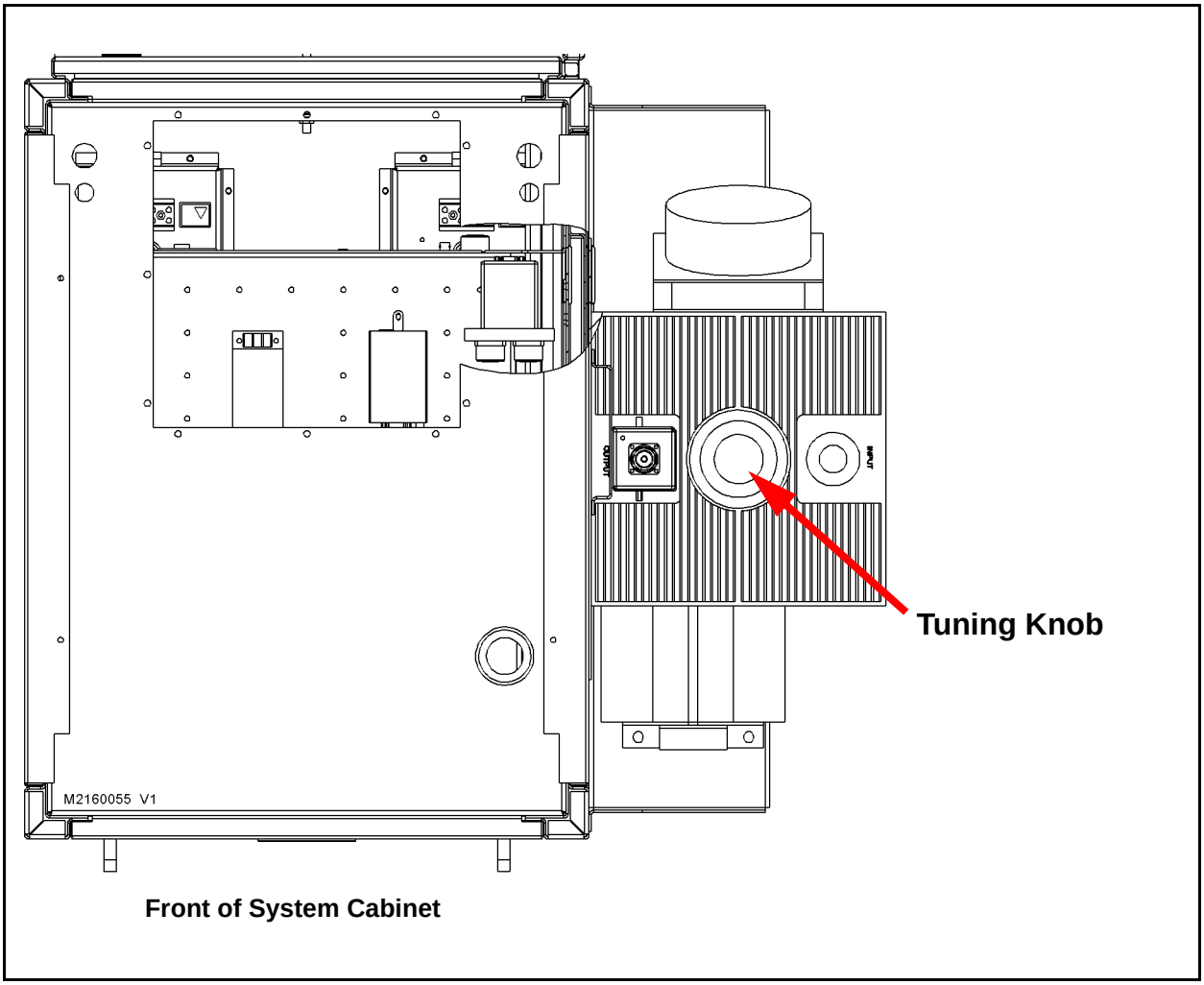
9. Disconnect the measurement cables from the bandpass filter and the RF output.

10. Reconnect coaxial cable connector W7P2 to the input of the bandpass filter (U2-INPUT).

11. Reconnect the antenna feed cable to the transmitter system's RF output (A4J2).

12. Proceed to "System turn-on" on page 4-8.

Figure 4.4: Tuning Knob Location



Tuning with an impedance meter

1. Turn off the transmitter system and disconnect (or disable) its ac power source(s).
2. Disconnect the P2 end of coaxial cable W7 from the input of the bandpass filter (U2-INPUT).
3. Disconnect the antenna feed cable from the transmitter system's RF output (A4J2).
4. Connect a calibrated 50 ohm termination (VSWR < 1.05:1) to the RF output (directional coupler output A4J2).
5. Loosen the brass locking nut at the bottom of the center tuning post using an 11/16" wrench.
6. Connect the measurement port of the impedance meter to the input of the bandpass filter (U2-INPUT). If the impedance meter requires calibration of its measurement port, perform the appropriate calibration.
7. Adjust the tuning knob (see Figure 4.4 on page 4-7) for minimum input VSWR (or maximum return loss).
8. Re-tighten the brass locking nut. Ensure the filter remains tuned for minimum input VSWR (or maximum return loss). Loosen and re-adjust if necessary.
9. Disconnect the impedance meter and remove the 50 ohm termination from the RF output.
10. Reconnect coaxial cable connector W7P2 to the input of the bandpass filter (U2-INPUT).
11. Reconnect the antenna feed cable to the transmitter system's RF output (A4J2).
12. Proceed to "System turn-on".

System turn-on

1. Reconnect (or enable) the transmitter system's ac power source(s).
2. On the system controller's front panel UI, go to the Control -> Channel Select screen. Set the channel frequency to the correct site frequency.
3. Press the system controller's RF On button.

Calibrating 10 MHz reference

If the measured system frequency is outside the desired limits and cannot be offset using the carrier trim feature (see "Trimming the carrier" on page 2-29), use the following procedure to calibrate the carrier frequency using the 10 MHz reference.

Test equipment and tools

- Frequency counter
- Coaxial cable assembly Nautel Part # 216-5120 (provided in ancillary kit)

System controller 10 MHz calibration

1. Connect a frequency counter to the **10MHz CAL IN/OUT** BNC connector (A1J8) on the rear of the system controller. A coaxial cable (Nautel Part # 216-5120) is provided in the ancillary kit.
2. On the system controller's front panel UI, go to the **Calibration** -> **Sys Con 10MHz Cal** screen (see "Sys Con 10MHz Cal" on page 2-40 of this manual). Adjust the calibration value until the frequency counter reads 10.000000 MHz.

Transmitter A and B 10 MHz calibration

1. Connect a cable between the **10MHz CAL** BNC connector (A1J8) on the rear of the system controller and the **10MHz CAL** BNC connector (A1J3) on the rear of transmitter A.
2. On the system controller's front panel UI, go to the **Calibration** -> **TXA 10MHz Cal** screen (see "TXA 10MHz Cal" on page 2-42 of this manual). Press the **SELECT** button to start the calibration. When the calibration routine is complete, press the **BACK** button.
3. Repeat Step 1 and Step 2 for transmitter B.

Section 5: List of terms

This section defines some of the terms that are used in Nautel documentation.

Cutback. A reduction in RF output power, caused by the occurrence of multiple shutbacks within a pre-defined period.

DSP. Digital Signal Processing.

EEPROM. Electrically Erasable Programmable Read-Only Memory.

Foldback. A reduction in RF output power, caused by adverse load conditions (high VSWR). No shutbacks or cutbacks have occurred.

Intermediate Power Amplifier (IPA). Refers to circuitry within the transmitter's RF power modules which amplifies the exciter's RF output to a level sufficient to drive the final RF amplifiers.

LED. Light Emitting Diode (also referred to as lamp).

PWB. Printed Wiring Board.

Shutback. A complete, but temporary loss of RF output power, caused by any one of a variety of faults, including high VSWR, high reject load power, RF drive failure, or an open external interlock.

Shutdown. A complete and permanent loss of RF output power. Typically follows repeated cutback, foldback or shutback events.

Surge protection panel. An electrical panel that protects equipment from electrical surges in the ac power supply, antenna or site ground caused by lightning strikes.

UI. The User Interface is the front panel display on the system controller that allows for extensive control and monitoring of the transmitter system.

VFD. Vacuum Fluorescent Display.

VSWR. Voltage standing wave ratio. This is an expression of the ratio of forward voltage to reverse voltage on the feedline and antenna system. An ideal VSWR of 1:1 provides maximum transmitter-antenna efficiency.

NG1000

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Nautel Limited
10089 Peggy's Cove Road
Hackett's Cove, NS Canada B3Z 3J4
Phone: +1.902.823.3900 or
Toll Free: +1.877.6NAUTEL (6628835) (Canada & USA only)
Fax: +1.902.823.3183

Nautel Inc.
201 Target Industrial Circle
Bangor, Maine USA 04401
Phone: +1.207.947.8200
Fax: +1.207.947.3693

Customer Service (24 hour support)
+1.877.628.8353 (Canada & USA only)
+1.902.823.5100 (International)

Email: support@nautel.com
Web: www.nautel.com

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